Not All That Glitters Is Gold: Firm Hiring in the Market for Knowledge Workers*

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Abstract

In frontier sectors such as biotechnology and artificial intelligence, firms increasingly compete to hire PhD-trained scientists to drive innovation. At the time of hire, firms face substantial uncertainty about how individuals will perform in industry, yet they can observe measures of academic productivity. How informative is academic performance about individuals' productivity in industry, and how does this impact hiring outcomes? Using confidential administrative data on 40 cohorts of PhD graduates across all fields linked to wage and publication records, I estimate individuals' productivity in both academia and industry, regardless of their actual sector of employment. I do so using a Marginal Treatment Effects framework that accounts for endogenous sorting into sectors and exploits variation in industry demand for PhDs at graduation. I find that academic productivity provides only limited information about individuals' productivity in industry, with an average correlation of 0.20–0.29. This informativeness has declined over time and varies markedly across majors. In majors where academic productivity is a weaker proxy for industrial productivity, firms' hires are significantly more likely to be laid off, consistent with greater difficulty identifying individuals whose skills translate effectively to industry. Together, the results suggest that limited alignment between academic and industrial productivity may create systematic challenges for firms when hiring scientific talent.

Keywords: *Hiring Under Uncertainty*, *Knowledge Workers*, *Talent Allocation*, *Institutional Environment*

*The use of NSF data does not imply NSF endorsement of the research methods or conclusions contained in this report. Any opinions, findings, conclusions, or recommendations expressed here do not reflect the views of NSF.

1 Introduction

Firms increasingly compete to hire scientific talent, particularly in knowledge-intensive sectors such as artificial intelligence and biotechnology (Hiraiwa et al., 2025). These individuals play a central role in shaping firms' innovative activities and are often embedded in projects that are costly to redirect or unwind (e.g., Henderson and Cockburn, 1994). Yet, firms face considerable uncertainty when hiring scientists. Their training and prior experience are typically rooted in academic environments, leaving their productivity in industry largely unobserved at the time of hire (Bidwell, 2011; Campbell, Coff and Kryscynski, 2012). As a result, firms often rely on readily observable academic records to identify potential hires, such as publications and other indicators of scientific performance (Nature Careers, 2025). Whether such information provides reliable guidance about who will perform well in industry, however, is not obvious.

Academia and industry are different institutional environments, characterized by distinct objectives, performance metrics, and incentives (Gittelman and Kogut, 2003; Kaiser et al., 2018; Merton, 1973; Stern, 2004). Academia rewards intellectual autonomy and novelty, while industry emphasizes execution under constraints and a focus on product-oriented outcomes (Bikard, 2018). Hence, while some skills such as technical expertise or creativity are valued in both sectors, others are more critical in one context than in the other. For example, the ability to work toward commercially valuable outcomes and meet deadlines is essential in industry but less emphasized in academic settings, where theoretical depth and independent inquiry are more important (Agarwal and Ohyama, 2013; Evans, 2010; Sauermann and Stephan, 2013). As a result, the scientists who would be the most productive in academia might not be those who would be the most productive in industry.

This can create challenges for firms seeking scientific talent. When academic productivity is an imperfect indicator of individuals' productivity in industry, academic records may provide only limited guidance for identifying which scientists will excel inside firms. Individuals who stand out in academic settings are not necessarily those who will perform best in industry, while others whose strengths are less visible in academic metrics may prove highly effective in firm-based work (Sauermann and Cohen, 2010; Stern, 2004). As a result, firms may hire scientists whose academic strengths do not translate to industry or overlook individuals with strong industrial potential, with consequences for organizational performance (Barney, 1991).

In this paper, I ask: *How informative is academic productivity about scientists' performance in industry? How does the informativeness of academic productivity shape firms' hiring outcomes?*

Conceptually, I characterize hiring as a problem shaped by context-dependent productivity. Individuals possess multiple skills whose value depends on the institutional environment in which they are deployed. As a result, each individual can be characterized by two potential productivity levels:

how productive they would be in academia and how productive they would be in industry, reflecting the interaction between their skill profile and the skills rewarded by each sector. The central object of interest is whether individuals' relative performance is preserved across contexts, namely, whether those who outperform others in academia also tend to outperform others in industry. I summarize this relationship using the correlation between individuals' productivity in academia and in industry, which captures the degree of alignment between sector-specific productivity and reflects how informative academic performance is about individuals' productivity in industry. This correlation depends on how skills are bundled within individuals and on the degree of overlap in the skills rewarded by academia and industry. Both dimensions are largely uncertain ex ante, rendering the relationship between productivity in academia and industry theoretically ambiguous.

To answer my research questions, I need to estimate each individual's productivity in both academia and industry, regardless of their actual sector of employment. This is empirically challenging for several reasons. First, it requires detailed longitudinal data on individuals' careers and measures of productivity. Second, because most individuals are observed working in only one sector, counterfactual productivity in the alternative sector is unobserved. Third, sector choice is endogenous: individuals self-select into academia or industry based on unobserved productivity components and preferences, which prevents recovery of counterfactual productivity using observed characteristics alone. Finally, addressing my research questions requires an empirical approach that separately recovers productivity in academia and productivity in industry, rather than only their difference (e.g., an average treatment effect).

I address these challenges using a confidential census of PhD graduates across all fields from U.S. universities between 1970 and 2013, with information about their employment at graduation. I link this dataset to multiple waves of a confidential administrative survey that track individuals over their careers, and use wages and publications to measure individuals' productivity (Bikard et al., 2019; Campbell, Ganco, Franco and Agarwal, 2012; Carnahan et al., 2012; Levin and Stephan, 1991). To recover each individual's productivity in both academia and industry, I estimate Marginal Treatment Effects (MTE) of sector choice on wages and publications. This method relies on an instrument to account for endogenous sorting. The first step estimates each individual's probability of entering industry based on observables and the instrument. Among individuals with similar predicted probabilities, some are observed in academia and others in industry. This variation allows me to trace academic and industrial productivity over the predicted probability of entering industry and, in turn, to recover each individual's productivity in both sectors. My instrument leverages variation in relative hiring opportunities between industry and academia at the time a PhD student graduates to generate plausibly exogenous variation in their likelihood of entering industry relative to academia (Arellano-Bover, 2024; Oyer, 2006).

I find that academic productivity is only weakly informative about scientists' productivity in industry. When productivity in academia and industry is proxied by individuals' estimated earnings in each sector, the correlation between academic and industrial productivity is 0.29. Using individuals' estimated publications in academia as an alternative measure of academic productivity yields an even weaker relationship, with a correlation of 0.20. This implies that individuals with below-median productivity in academia still have a 40% probability of having above-median productivity in industry, highlighting the substantial uncertainty firms face when extrapolating from academic performance. The relationship between academic and industrial productivity has also changed over time. Between 1980 and 2010, the correlation declined by 34%, indicating that academic productivity has become a less reliable predictor of industrial productivity for more recent cohorts. This pattern is consistent with broader changes in the organization of industrial R&D, including evidence that firms have shifted away from upstream scientific research toward downstream development and application (Arora et al., 2018). Finally, I document substantial heterogeneity across PhD majors. Correlations range from -0.04 to 0.58, implying that academic productivity is strongly predictive of industry performance in some domains, but misleading in others.

I next show that weaker alignment between academic and industrial productivity translates into worse hiring outcomes for firms. To that end, I use layoffs as a revealed measure of post-hire underperformance and exploit variation across PhD majors in the correlation between academic and industrial productivity. I find that industry hires are more likely to be laid off in majors where this correlation is lower: a 0.1 decrease in the correlation is associated with a 15% increase in the layoff rate relative to the mean. While descriptive in nature, this result is consistent with an evaluation challenge with direct implications for hiring outcomes: when academic productivity is a less informative proxy for industrial productivity, academic records become less effective screening tools, increasing the likelihood that firms hire individuals who subsequently underperform. Importantly, these effects are not uniform across firms or roles. Heterogeneity analyzes suggest that these evaluation challenges are more pronounced for smaller firms and R&D occupations. Smaller firms are likely to have more limited screening capacity and fewer opportunities for internal redeployment, while R&D positions place greater weight on academic productivity when hiring individuals. As a result, hiring outcomes in these settings are especially sensitive to variation in how informative academic productivity is about individuals' performance in industry.

This paper makes several contributions. First, it advances our understanding of hiring under uncertainty by showing that relative performance across individuals is not necessarily preserved across institutional contexts. Much prior work implicitly assumes that workers possess a single, context-invariant ability that preserves their relative performance across settings, an assumption often operationalized through worker fixed-effects or general performance controls (Agarwal and

Ohyama, 2013; Bhaskarabhatla et al., 2021; Kim, 2018; Roach and Sauermann, 2024). I relax this assumption by directly estimating individuals' productivity in academia and in industry, allowing me to characterize the relationship between academic and industrial productivity (Bikard, 2018; Dasgupta and David, 1994; Sauermann and Roach, 2014; Stephan, 2012). I further show that this relationship varies systematically across PhD majors and has changed over time, identifying clear boundary conditions under which academic performance is more or less informative about individuals' productivity in industry. By focusing on whether the *same* individuals perform well across sectors, this paper complements prior work on firm- and team-specific human capital (Groysberg et al., 2008; Jaravel et al., 2018) which often documents a decline in individuals' performance following mobility.

This paper also contributes to the innovation and strategy literature on how knowledge workers shape firm performance by linking cross-context differences in the informativeness of academic productivity to firms' hiring outcomes (Baruffaldi and Poege, 2025; Gambardella et al., 2015; Palomeras and Melero, 2010). Prior work has often treated scientists as an homogeneous input into firms' R&D processes (Arora et al., 2024, 2025) or examined their productivity within a single institutional context (Azoulay et al., 2017, 2011; Ganco, 2013; Ganco et al., 2015; Oettl, 2012). As a result, we know comparatively little about how firms' evaluation of scientists translates into downstream outcomes (Agrawal, 2001). I show that variation in the informativeness of academic productivity has direct consequences for firms' hiring outcomes: when academic productivity is a weaker proxy for industrial performance, firms are significantly more likely to hire individuals who underperform and are subsequently laid off. This highlights an important strategic implication: evaluating individuals based on performance indicators developed in a different institutional context can undermine the quality of firms' hiring outcomes. This finding also underscores that leveraging scientific talent is not simply a matter of acquiring highly trained individuals, but rather of identifying the subset whose skills align with the private sector's institutional environment (Arora and Gambardella, 1994; Cohen and Levinthal, 1990; Tranchero, 2023).

Methodologically, I use a Marginal Treatment Effects framework that allows the recovery of individual-specific counterfactual productivity in both academia and industry (Brinch et al., 2017; Heckman and Vytlacil, 2007). This econometric approach remains relatively uncommon in the management literature, but is uniquely suited to capture individual-level heterogeneity. This helps answer recent calls to move beyond average effects to better capture the sources of heterogeneity giving rise to organizations' competitive advantage (Coff and Rickley, 2021; Snyder et al., 2024).

The paper proceeds as follows. Section 2 introduces the conceptual framework. Section 3 presents the estimation method. Section 4 discusses the empirical setting, data and measurement strategy. Section 5 describes the research design. Section 6 reports the results. Section 7 shows additional robustness analyses and Section 8 concludes.

2 Conceptual Framework

In this section, I develop a framework where productivity is individual-specific and context-dependent, arising from the interaction between individuals' skills and the context in which they are deployed. The framework focuses on the relationship between individuals' productivity in academia and in industry, and on the implications of this relationship for hiring outcomes.

2.1 Hiring Under Uncertainty About Individuals' Productivity

Human capital is widely recognized as a key source of competitive advantage, particularly in knowledge-intensive sectors where individuals' technical expertise, creativity and domain-specific tacit knowledge drive organizational outcomes (Barney, 1991; Coff and Kryscynski, 2011; Coff, 1997). A central insight from this literature is that productivity is not solely a function of individuals' skills, but also of how those skills interact with the organization itself (Lazear, 2009). From this perspective, individuals represent bundles of skills accumulated through education, experience, and innate ability, whose value depends on the organizational context in which they are deployed. That is, the value of human capital is both individual-specific and context-dependent: the same person may be highly productive in one setting and less so in another. As a consequence, hiring individuals whose skills align with the organization's unique environment is the foundational first step in building a valuable workforce that can generate competitive advantage (Kryscynski, 2021).

Hiring, however, takes place under substantial uncertainty. Firms must identify individuals who are likely to generate value within their own organizational environments (Cappelli, 2019; Huang and Cappelli, 2010), yet at the time of hire they typically lack direct information about how workers' skills will translate into productivity, particularly for external hires (Bidwell, 2011; Coff and Kryscynski, 2011). As such, firms commonly rely on indicators, such as credentials or prior roles, that reflect individuals' performance in *other* institutional environments, often governed by different objectives and logics. As a result, a key question is whether performance in one institutional context helps identify individuals who will be productive in another. When individuals who perform better than others in one setting also tend to perform better than others in a different setting, indicators generated in the first environment provide useful guidance for hiring. When relative performance is not preserved across contexts, however, these same indicators can be weak or misleading.

The boundary between academia and industry provides a particularly well-suited setting to study hiring under uncertainty across contexts. First, the two sectors are governed by distinct institutional logics, yet they draw from a common labor pool, with firms frequently hiring scientists directly out of academia. Moreover, firms face substantial uncertainty about individuals' productivity in industry, as scientists hired from academic environments typically have little to no industrial experience. Instead, firms can observe standardized measures of individuals' productivity generated in academia, such as publications, citations, or institutional affiliation. Finally, hiring scientists is costly and difficult to reverse: scientists are embedded in long-horizon projects, integrated into teams and pipelines, and evaluated over extended periods, limiting firms' ability to rely on trial-and-error hiring. Together, these features make the academia–industry boundary a setting in which the challenges of evaluating talent across contexts are particularly acute.

2.2 A Simple Framework for Hiring Across Contexts

To formalize the hiring problem described above, consider two sectors of employment, academia and industry, indexed by $j \in \{A, I\}$. Each sector j values a set of skills k^j. Individuals are indexed by i and possess a certain level of skills valued by each sector, k_i^j. Each individual is therefore characterized by a pair of potential outcomes: how productive they would be in academia P_i^A, and how productive they would be in industry P_i^I.

At the time of hire, firms have limited information about individuals' productivity in industry P_i^I. Instead, they might observe outcomes generated in academic environments, such as publications, citations, or institutional affiliations, that capture information about individuals' productivity in academia, P_i^A. The extent to which individuals' productivity in academia is informative about their productivity in industry hinges on whether individuals who are relatively more productive than others in academia also tend to be relatively more productive than others in industry. Importantly, this is a question about relative productivity across contexts, rather than about absolute productivity levels. Even if all individuals are more or less productive in one context than the other, academic productivity is useful for hiring to the extent that individuals who are more productive than others in academia also tend to be more productive than others in industry.

I refer to the relationship between productivity across contexts as the *alignment* between individuals' productivity in academia and individuals' productivity in industry, and I summarize it using the correlation between sector-specific productivity: $Corr(P_i^A, P_i^I)$. Figure 1 illustrates three stylized cases. A positive correlation (Figure 1a) implies that relative productivity is preserved across contexts, so that academic productivity provides meaningful information about individuals' productivity in industry. A negative correlation (Figure 1b) implies that academic productivity is systematically misleading. A weak or null correlation (Figure 1c) implies that academic productivity offers little guidance for distinguishing individuals by their industrial potential. As the correlation between productivity in academia and industry weakens, academic performance becomes a less reliable screening device, and firms face greater difficulty identifying candidates whose skills will translate effectively to industry.

2.3 The Institutional Environments of Industry and Academia

The correlation between individuals' productivity in academia and in industry is shaped by: i) the overlap in the skills each sector rewards ii) how those skills are distributed across individuals. Productivity across contexts will tend to be positively related when the two sectors reward a common foundation of skills, or when they reward different skills but individuals typically possess both. By contrast, individuals' productivity in academia and in industry will be negatively related when these sectors reward distinct skills and individuals possess one set of skills but not the other. In Appendix C, I formally model the correlation between academic and industrial productivity. Here, I summarize the key intuition and discuss, in turn, the skills valued by academia and industry and the structure of skill endowments across individuals.

Knowledge workers in both academia and industry, particularly for research and development roles, are typically expected to contribute to innovation. As a result, both sectors are likely to value a common set of foundational skills such as analytical reasoning, technical depth, and creativity, which support problem-solving and innovation in a range of settings (Sauermann and Roach, 2014; Stern, 2004). However, each sector has its own objectives, evaluation criteria, and incentive systems, which shape how productivity is defined and what skills are rewarded (Gittelman and Kogut, 2003; Perkmann et al., 2013; Stuart and Ding, 2006; Tartari and Breschi, 2012).

The academic reward system is anchored in reputation-based evaluation and open dissemination of knowledge. These features are reinforced by a broader set of scientific norms - universalism, communism, disinterestedness, and organized skepticism - that structure behavior and status within academic communities (Dasgupta and David, 1994). As such, the prevailing academic logic emphasizes intellectual autonomy, peer recognition, and theoretical novelty. Researchers are expected to pursue independent inquiry and produce original ideas (Hill and Stein, 2025; Merton, 1973). These incentives reward skills such as originality, self-direction, deep expertise, and the ability to frame contributions in ways that are compelling to scholarly audiences.

In contrast, industry emphasizes restricted disclosure, and the generation of knowledge that can be privately appropriated and applied toward commercial ends. Knowledge workers in firms face time and execution constraints and are expected to contribute to product development, process improvements, and commercially valuable outcomes (Agarwal and Ohyama, 2013; Bikard, 2018; Sauermann and Stephan, 2013). This institutional context rewards skills such as cross-functional teamwork, strategic prioritization, responsiveness to feedback, and the ability to integrate scientific work into larger project pipelines or commercial goals, under organizational and time constraints. As one executive said in my interviews: *"We don't need brilliant thinkers, we need scientists who can move fast, handle ambiguity, and deliver results."* Recent work has also highlighted a structural shift: firms have moved away from upstream scientific research toward downstream development and application (Arora et al., 2018). As firms reduce their engagement with basic science, the skills required for success in corporate R&D may become increasingly distinct from those fostered in academic environments. Overall, it remains unclear how much overlap exists in the skills each sector rewards.

On the supply-side, the relationship between academic and industrial productivity also depends on how skills are distributed across individuals. Even when sectors value distinct skills, academic and industrial productivity may still be positively correlated if individuals tend to possess both types of skills. Sector-specific skills need not be mutually exclusive: individuals may be naturally endowed with both sector-specific skills or develop them in parallel. While direct evidence on skill co-occurrence is limited, prior work suggests that scientists often make strategic trade-offs based on what their environment rewards (Lazear, 2009; Sauermann and Roach, 2014; Stern, 2004). As another executive noted in my interviews: *"The best scientist is not always the best drug hunter. Some scientists are excellent at pushing data further and extracting publishable insights, which is valuable in academia. But they lack what matters in drug discovery: the ability to evaluate risk and work with incomplete data, to make strategic judgments, and to provide a clear recommendation."*

In sum, the correlation between individuals' productivity in academia and in industry is theoretically ambiguous. This ambiguity reflects how different institutional environments value different skills, and the limited evidence on how skills co-occur or trade off across individuals. Empirically estimating the relationship between academic and industrial productivity is therefore essential to examine the challenges firms may face when hiring scientific talent.

3 Empirical Framework

3.1 Parameters of Interest and Empirical Challenges

Estimating the relationship between individuals' productivity in academia and in industry requires me to separately estimate P_i^A and P_i^I. There are three main challenges in doing so. First, individuals join one of the two sectors, so I only observe their productivity in academia or in industry.[1] Second,

[1]Some individuals switch sector during their career, allowing productivity to be observed in both academia and industry. However, such movers constitute a small and highly selected subset of individuals, as sector switching is likely endogenous to individuals' potential productivity in both sectors. As a result, observed productivity for movers reflects selection and post-switch adjustment rather than the counterfactual productivity that would characterize the

I cannot simply recover an individual's counterfactual productivity in the sector that they did not join by making inferences based on observable characteristics because sector of employment is endogenous: individuals sort into academia or industry based on unobserved factors, such as preferences or comparative advantage, that may be correlated with productivity in both contexts.[2] Third, standard empirical approaches such as differences-in-differences or two-stage least squares are designed to recover treatment effects (i.e. $P_i^I - P_i^A$), but do not separately identify the two potential productivity parameters needed to characterize the correlation between P_i^A and P_i^I.

3.2 Estimation Technique

A natural benchmark approach in the presence of endogenous sector choice is Two-Stage Least Squares (TSLS) using an instrumental variable. However, TSLS is not well suited for the objectives of this paper: (i) TSLS estimates a *difference* between two potential outcomes (i.e., $P_i^I - P_i^A$), rather than the two potential outcomes separately, and (ii) TSLS identifies this effect only for a local subset of individuals - the compliers - rather than across the full distribution of individuals. To recover estimates of P_i^A and P_i^I separately as functions of observed characteristics and unobserved heterogeneity, I estimate instead Marginal Treatment Effects (MTE) (Björklund and Moffitt, 1987; Cornelissen et al., 2016; Heckman and Vytlacil, 1999, 2007). MTE also relies on the use of an instrument, but provides a more general framework than TSLS. I present here the main intuition behind the MTE estimation and refer the reader to Appendix E for a more comprehensive discussion.

The estimation of MTE relies on the same key ingredients as TSLS: i) an outcome (in this paper, productivity) ii) an independent variable (joining industry)[3] and iii) an instrument. The first step of the MTE estimation is similar to the first stage of a TSLS: I estimate the probability that individual i enters industry as a function of observed covariates X_i and an instrument Z_i. This yields a propensity score, $P(Z_i, X_i)$, which summarizes the predicted likelihood that individual i joins industry given their observables and the instrument.

The central insight of MTE is that, conditional on the same propensity score, some individuals are observed in industry and others in academia. This is because individuals differ in unobserved ways denoted by U_{D_i}, such as preferences or comparative advantage, which influence sector choice even among individuals with the same predicted likelihood of entering industry based on X_i and Z_i.[4]

broader population. In contrast, my approach exploits variation in initial sector choice to recover potential productivity in both sectors for the full population of PhD graduates, including those who never switch sectors.

[2]See Appendix D for details.

[3]I use sector joined *at graduation* because it allows me to estimate productivity trajectories from the beginning of individuals' careers. While the analysis is anchored at entry, the framework has implications for hiring decisions beyond initial career placement.

[4]U_{D_i} is often called 'unobserved resistance' or distaste for treatment and is assumed to be uncorrelated with the instrument itself.

For a given value of the propensity score $P(Z_i, X_i) = p$, individuals observed in academia must have relatively high values of U_{D_i} (a stronger inclination toward academia), while those observed in industry must have relatively low values of U_{D_i}. As a result, the average observed productivity of individuals in academia at propensity score p is informative about productivity potential in academia for individuals with $U_{D_i} \geq p$, while the average observed productivity of individuals in industry at the same propensity score is informative about productivity potential in industry for individuals with $U_{D_i} \leq p$.[5] By modeling observed productivity as smooth functions of the propensity score separately within academia and industry, the MTE framework recovers productivity potential in academia and in industry as a function of observed characteristics and unobserved heterogeneity, i.e., $E[P_i^A | X_i = x, U_{D_i} = u]$ and $E[P_i^I | X_i = x, U_{D_i} = u]$.

Intuitively, if an individual's position in the unobserved heterogeneity distribution was known, their potential productivity in each sector could be inferred. While U_{D_i} is not directly observed, the propensity score and realized sector choice jointly provide information about an individual's location in this distribution. This mapping allows me to generate individual-specific predictions for individuals' productivity in both academia and industry, conditional on observed characteristics, the propensity score, and sector choice, i.e., $E[P_i^A | X_i, D_i, p_i]$ and $E[P_i^I | X_i, D_i, p_i]$.

The identifying assumptions underlying the MTE framework are the same as those required for TSLS: instrument relevance, exclusion, and monotonicity. In the absence of additional structure, nonparametric identification of MTE would require full support of the propensity score for individuals observed in both academia and industry at all values of the observed covariates X, a condition rarely satisfied in practice (Andresen, 2018). Following the literature, I also impose additive separability between observable and non-observable components of productivity. This allows identification of the MTE over the region of common support of the propensity score, without requiring full support conditional on X (Carneiro et al., 2011; Cornelissen et al., 2016).

4 Setting, Data and Measurement

4.1 Data

I use individual-level data from three confidential files: (i) the Survey of Earned Doctorates (SED), (ii) the Surveys of Doctorate Recipients (SDR) and (iii) the 2015 Survey of Doctorate Recipients Bibliometric Research Data (SDR15-WoS). These surveys are conducted by the National Center for Science and Engineering Statistics (NCSES) and cover individuals who received a PhD from

[5]Formally, these correspond to $E[P_i^A | X_i = x, U_{D_i} \geq p]$ and $E[P_i^I | X_i = x, U_{D_i} \leq p]$.

an American institution.[6]

The SED is an *annual census* of all individuals who graduated from a PhD in a given academic year, conducted since 1958 (e.g., the SED 2010 is sent to all individuals who earned their doctoral degree between 1 July 2009 and 30 June 2010).[7] The response rate is about 90%, providing almost full coverage of the population of individuals who receive their PhD in the United States in a given year. The SED contains information about doctoral studies (e.g., PhD field of study and institution) as well as information about previous education (e.g., bachelor and master). It also contains demographics and background information (e.g., place of birth, gender, race, parental education, citizenship status). Importantly for the analysis, the SED contains information about postgraduation plans (e.g., work, postdoc, other study or training).

The SDR is a *biennial panel survey* of doctorate holders conducted since 1973 which contains longitudinal data about individuals' employment at the time of survey - i.e., during their career - such as sector of employment, principal job activity and earnings. For each survey, the sample consists of both individuals who were surveyed in previous SDR editions (as long as they are less than 76 years old) as well as new doctoral graduates who earned their PhD since the last cycle. The response rate is about 65%. Individuals can be surveyed several times during their career and hence might appear in several SDR waves. While the SDR reports institution name for individuals who work in academia, I do not have employer identifier for those who don't.

In the years 2022/2023, the NCSES launched a new Research Data Infrastructure with the goal of creating new linkages with external data. I take advantage of this effort by using a newly created dataset that matches each individual surveyed in the SDR 2015 to their research output from graduation until 2017, using information from Web of Science. This gives me individual-level information about publication output produced between graduation and 2017 such as number of publications, top publications and average number of citations in the first 2 or 5 years after publication. For each of these measures and for each individual, I have access to aggregate values up to 2017 as well as more granular information by bins of experience (e.g., publications produced during the PhD, publications produced during the first 5 years post-graduation and so on).[8] Note that I do not have information at the publication-level such as journal or abstract.

I first match the SDR 2015 to the SDR15-WoS and the SED. This gives me information about all individuals surveyed during their career in 2015 regarding their background, doctoral education,

[6] These data are confidential and were accessed remotely through the National Opinion Research Center (NORC) Data Enclave under a restricted-use license approved by the National Science Foundation (NSF).

[7] For reference, about 55k research doctorates were awarded in the United States in 2020. This excludes recipients of professional doctoral degrees, such as MD, DDS, DVM, JD, DPharm, DMin, and PsyD.

[8] I have counts of these measures for the 5 years before graduation and for bins of 5 years after graduation, starting for bin of years 1 to 5 after graduation up to bin of years 46 to 50 after graduation.

employment plans at graduation, employment status in 2015, earnings in 2015, as well as publication output. I then match this sample to the other SDR surveys I have access to (survey years 1995, 1997, 1999, 2001, 2003, 2006, 2008, 2010, 2013, 2017 and 2019). This allows me to retrieve additional observations about individuals' earnings during their career.

I restrict the analysis to Science and Engineering PhD majors and exclude Health-related fields, whose graduates typically follow clinical career paths. I further exclude cohorts graduating before 1970 due to changes in survey coding. To identify individuals' initial sector of employment, I retain those with a definite postdoctoral commitment at graduation and exclude internships, residencies, traineeships, and military service. I also restrict the sample to individuals who start their careers in the United States. Additional exclusions reflect data requirements for identification and estimation. I exclude individuals with missing information on post-graduation employment, the instrumental variable, earnings, or any covariates included in the regressions. Finally, because the estimation requires observing at least one realized productivity outcome in an individual's initial sector, I exclude individuals who are never observed working in their starting sector. Table B.1 provides a detailed sample construction table reporting all exclusions and sample sizes at each step.

The resulting sample includes 22,609 unique individuals. Because the earnings outcome is a flow and doctorate holders can be surveyed several times during their career, my sample contains 69,917 observations in total, meaning that I observe on average 3 earnings values per individual.

I refer to PhD *major* as the most granular field of study I have access to for each individual (e.g., Biology, Genetics, Biochemistry, Astronomy, Computer Engineering etc.). This is what students choose to study. My dataset includes 58 distinct PhD majors. I classify these majors into 5 PhD *fields*: Computer Sciences and Mathematics, Life Sciences, Physical Sciences, Social Sciences and Engineering (see Figure A.1 for a list of PhD fields and majors).

4.2 Dependent Variables: Measurement of Productivity

From a theoretical perspective, the ideal measure of an individual's productivity is their marginal product of labor, i.e., the value they create for the organization. For knowledge workers, however, such marginal contributions are rarely directly observable. As a result, empirical work typically relies on proxies that aggregate information about individuals' contributions. The most widely used proxy for productivity in this literature is earnings, which reflect how organizations value individuals' overall contributions (Campbell, Ganco, Franco and Agarwal, 2012; Rosen, 1981).

Accordingly, I estimate individuals' potential earnings in academia and in industry as proxies for their sector-specific productivity. One concern is that earnings in academia are often shaped by institutional pay scales and tenure-based salary progression, which may compress wages by limiting variation across individuals.[9] Therefore, I also use data on research output and estimate individuals' potential publication output in academia as a complementary measure of academic productivity. While earnings allows me to capture overall contributions, including teaching, administration, mentoring, and service, publication output primarily captures research productivity.

Earnings: Earnings correspond to the total earned income before deductions in the previous year. This includes wages, bonuses, overtime, consulting fees and summertime teaching or research, and I have access to several earnings observations per individual at different years of experience.[10] I transform all values in 2015 US dollars. I winsorize observations below the 1^{st} percentile and above the 99^{th} percentile to limit the influence of extreme values.

Publications: I calculate the number of publications published after PhD graduation by subtracting the number of publications published before graduation from the total number of publications published by the individual until 2017.[11] I winsorize observations below the 1^{st} percentile and above the 99^{th} percentile. Depending on the specification, I also use publications weighted with citations received in the first 5 years of publication.

Patents: In the 1995, 2001, 2003 and 2008 survey waves, respondents could report whether they have been named as an inventor on any application for a U.S. patent and if so, how many applications list them. I use this information to estimate individuals' patenting output in industry as another proxy for individuals' productivity in industry. Coverage is limited: only 3,599 individuals in my sample report patenting information. Hence, I primarily rely on this variable as robustness.

4.3 Independent Variable: Measurement of Sector Joined

My main independent variable of interest is the sector that individuals join upon graduation. I first assign each individual to three potential sectors: industry (business for-profit and self-employed), academia (2-year and 4-year college or university and university-affiliated research institute) or government (US or foreign government). My preferred treatment variable is an indicator variable equal to 1 for individuals joining industry and 0 for individuals joining any other sector, i.e., academia or government (in the rest of the paper, I will loosely refer to this treatment variable as joining industry vs academia).[12] Overall, among the 22,609 unique PhD graduates in my sample,

[9]Because my goal is to estimate relative productivity within each sector, what matters empirically is the rank ordering of individuals, not absolute productivity levels. While wage compression could attenuate observed differences, it is unlikely to meaningfully distort rank-ordering. Moreover, academic earnings correlate positively with other indicators of productivity, such as publication counts, citation impact, and research funding (Hilmer et al., 2015; Katz, 1973).

[10]To avoid contaminating the estimation of P_i^A and P_i^I, I exclude earnings observations that occur after individuals switch sectors - that is, earnings corresponding to the sector not initially chosen. For example, using earnings in industry from someone who initially joined academia would improperly enter into the estimation of P_i^A.

[11]I only include publications that were articles, reviews or conference proceedings.

[12]I show in Section 7 that results are robust to excluding the government category (10% of the sample).

21% are joining industry upon graduation.

4.4 Descriptive Statistics

Figure A.2 shows the number of PhDs by cohort of graduation and PhD field in my sample. The number of individuals increases with cohort of graduation, reflecting the increase in the number of PhDs over time. Figure A.3 shows the share of PhDs joining the private sector upon graduation by PhD cohort, which oscillates between 20% and 30%. There is, however, substantial heterogeneity across fields as shown in Figure 2: about 55% of PhDs in Engineering join the private sector, vs 10% in Life Science or Social Sciences.

Table 1 reports summary statistics at the individual level, by sector joined. Individuals who join the private sector earn on average across the career USD 67,000 more per year than individuals who started their career in academia. In a simple regression of logged earnings on sector of employment, controlling for observable characteristics, Table B.2 shows that the earnings premium in industry is about 40%. Individuals who join industry also publish on average 16 fewer publications between graduation and the time of survey. In academia, 44% of individuals are female, vs only 29% in industry. Overall, 72% of individuals in my sample are Americans but a higher share of international students tend to join the private sector at graduation than academia.[13]

Figure A.4 shows earnings as a function of experience and sector joined. The earnings curve in industry has a higher intercept but is less steep, so that earnings in both sectors look roughly similar in the raw data towards the end of the career (Levin and Stephan, 1991). Figure A.5 shows citation-weighted publications as a function of sector joined and experience. The curve is upward sloping in academia and noisier in industry, likely because individuals' activity in industry is less homogeneous, with workers i) not necessarily employed in research and/or ii) working in firms that do not allow them to publish research results.

5 Research Design

Estimating individuals' potential productivity in academia, P_i^A, and in industry, P_i^I, using the Marginal Treatment Effects framework described in Section 3 requires an instrumental variable that generates exogenous variation in individuals' sector of employment. The key endogeneity concern

[13] I follow the classification of the SED and identify U.S. citizens as individuals who are either U.S. native born (68% of my sample), U.S. naturalized (4% of my sample), U.S. unspecified (not present in my sample) or U.S. citizen assumed for Puerto Ricans who did not report citizenship status (not present in my sample). Individuals who are not identified as U.S. citizens can be non-U.S. immigrant permanent resident (5%), non-U.S. non-immigrant temporary resident (23%) or non-U.S. visa status unknown (0.1%).

is that individuals are likely to self-select into academia or industry based on comparative advantage and preferences, which may be correlated with their productivity in each sector. To address this concern, my empirical strategy exploits variation in relative hiring opportunities between industry and academia at the time individuals graduate from their PhDs.

Variation: The set of employment opportunities available to PhD graduates fluctuates across years due to factors such as technological advances and shifts in research priorities in both industry and academia. For instance, breakthroughs in gene-editing technologies like CRISPR can expand industry hiring opportunities for PhDs in genetics and biotechnology, as firms seek to explore new applications in agriculture and medicine. Similarly, universities often adjust their hiring based on evolving hiring needs, academic priorities, funding availability, and strategic research initiatives (Goolsbee and Syverson, 2023). As a result, PhD students in a given major face very different entry opportunities depending on the year they graduate. My strategy exploits individuals' timing of graduation to capture this underlying variation in relative hiring opportunities between industry and academia.

IV construction: Empirically, the instrument can be thought of as a vector of PhD major $\times$ PhD cohort fixed-effects, controlling for PhD major fixed-effects (when does someone graduate given their major). In practice, I create a leave-one-out continuous equivalent instead of the interaction. Excluding the individual's own observation ensures that the instrument is not mechanically influenced by the individual's sector choice and mitigates concerns about reflection bias (Sampat and Williams, 2019). For each individual, the instrument Z_i is:

$$Z_i = \frac{\sum_{i'} industry_{i'} \times \mathbf{1}\{m(i') = m(i)\} \times \mathbf{1}\{c(i') = c(i)\} - industry_i}{\sum_{i'} \mathbf{1}\{m(i') = m(i)\} \times \mathbf{1}\{c(i') = c(i)\} - 1} \tag{1}$$

with i' indexing individuals, $industry_i$ an indicator equal to 1 if individual i joins industry and 0 otherwise, $m(i)$ individual i's PhD major and $c(i)$ individual i's PhD cohort of graduation. This approach follows Oyer (2006) and, more recently, Arellano-Bover (2024).[14]

In this setting, the instrument is constructed from revealed cohort-level patterns of sector entry and therefore aggregates a range of underlying forces that shape the employment opportunities facing PhD graduates at the time of graduation. For identification, the relevant source of variation is differences across cohorts in the relative availability of positions in academia versus industry, which is plausibly exogenous from the perspective of PhD graduates. However, cohort-level entry patterns could also reflect other forces, such as individuals' choices of when to graduate, which could violate the validity of the instrument. I discuss potential threats to the instrument below considering

[14]I follow the literature and only keep PhD major $\times$ PhD cohort cells with at least 10 observations (Heckman, 1981).

earnings as the outcome of interest, but the discussion remains similar using publications.[15]

Exclusion restriction: A potential concern is that cohort-level variation in sector entry may also reflect changes in relative compensation across sectors, rather than solely differences in hiring opportunities. In that case, higher values of the instrument would also be associated with higher industry wages. Because earnings during the career are used as proxies for productivity, such price-driven variation would confound productivity with pay conditions.[16] To mitigate this concern, I control for PhD cohort fixed effects to absorb economy-wide wage movements that are common across sectors. A remaining concern is that field-specific technological or market shocks at the time of graduation could directly affect compensation within academia or within industry. I therefore also control for field-by-cohort higher-education R&D expenditures in the graduation year, separately for federally funded and business-funded R&D performed at universities.[17] Federal R&D expenditures proxy for field-specific technological and scientific shocks that may influence academic labor markets, while business-funded R&D expenditures capture field-specific market and commercial demand that could affect industry wages.[18]

In addition, I examine whether the instrument is correlated with within-sector changes such as shifts in employer or job characteristics that could also correlate with earnings. For example, if periods of strong industry entry are concentrated among large, high-paying firms, the instrument would affect wages within industry (not only the probability of entering industry). While the exclusion restriction cannot be proven, I implement several falsification tests, reported in Table B.3, and show that the instrument is not correlated with employer size or job occupation.

Independence assumption: The independence assumption requires that graduation timing is not systematically related to individuals' underlying skills or preferences. In particular, this implies that individuals who would be highly productive in industry, and who would earn more there, do not systematically graduate in years where industry hiring opportunities are relatively more prevalent. This assumption could be violated through the following channels, which I examine below: strategic graduation; strategic entry; skill composition shifts.

First, one concern could be that individuals who would be highly productive in industry strategically adjust their graduation year based on industry hiring opportunities at graduation. For instance, if industry hiring opportunities are relatively weak in a given year, such individuals might delay

[15] Figure A.6 formalizes how these assumptions relate to key variables in a simple model.

[16] Because the instrument operates at the time of entry, any price-driven component would most plausibly affect wages at entry. Focusing on medium- to long-term earnings therefore reduces concerns related to short-run wage premia, although persistent within-sector compensation differences remain a potential threat.

[17] Source: `https://ncses.nsf.gov/surveys/higher-education-research-development`

[18] In practice, higher R&D expenditures likely capture both increased wage pressure and increased hiring. Controlling for it may therefore absorb some relevant variation, making identification more conservative.

graduation in anticipation of more favorable conditions the following year. Conversely, individuals who would otherwise graduate later might accelerate graduation to take advantage of favorable entry conditions. To test for this threat, I use information about PhD entry year to calculate PhD duration and regress this variable on the instrument. Table 2 Column (1) shows no significant relationship between PhD duration and the instrument. To account for offsetting behaviors, I also estimate two additional specifications using indicators for whether an individual's PhD duration is strictly shorter or strictly longer than the average duration for individuals with the same PhD major and entry year. Columns (2) and (3) show no significant correlation. Overall, these results suggest that individuals do not systematically manipulate their graduation timing in response to cohort-level hiring opportunities.

Another concern is that individuals with high industry potential may be able to forecast future hiring opportunities several years in advance and strategically time their PhD entry. To assess this possibility, I regress the value of the instrument at graduation on the value of the instrument the year before PhD entry, since individuals apply to PhD programs usually the year before starting their program. Table 2 Column (4) shows no significant correlation, suggesting that hiring opportunities at graduation are not predictable based on information available at the time of PhD application. I also check whether market conditions the year before entry predict future sorting and find no significant effect in Column (5).

Third, gradual shifts in skill composition across PhD cohorts could correlate with changes in hiring opportunities between industry and academia. For example, sustained growth in industry hiring opportunities in certain fields could attract individuals with stronger industry-oriented skills, altering the average skill composition of graduating cohorts over time. I show in Section 7 that my results are robust to the inclusion of ten-year-window fixed-effects interacted with PhD field. This allows me to compare individuals graduating in the same decade and mitigate the risk of long-run shifts in skill composition.

Finally, students from different doctoral institutions may have systematically different career prospects or preferences, which could affect both their responsiveness to hiring opportunities and their subsequent earnings. To mitigate this concern, I include doctoral institution fixed-effects, along with fixed-effects for gender, race, birthplace, citizenship, and parental education. Overall, any remaining threat to the independence assumption would require unobserved, time-varying shifts in the composition of PhD graduates within a major that are correlated with fluctuations in relative hiring opportunities and not captured by my existing controls or my previous tests.

6 Results

6.1 First-Stage

Individuals' propensity score, defined as the probability that individual i enters industry conditional on observables and the instrument, is estimated with the following regression:

$$industry_i = \gamma_0 + \gamma_1 Z_i + \gamma_2 \mathbf{X_i} + \nu_i \tag{2}$$

where $industry_i$ is an indicator equal to 1 if individual i joins industry and 0 otherwise. Z_i is the instrument for individual i, which is calculated at the PhD major $\times$ PhD cohort level. Following my identification strategy, $\mathbf{X_i}$ includes PhD major fixed-effects, PhD cohort fixed-effects, as well as the log of field-specific federal and private R&D funding at the time of graduation. $\mathbf{X_i}$ also includes a linear and quadratic terms to control for experience as I observe individuals at different points in time during their career, an indicator equal to 1 if the individual declares being female and 0 otherwise, an indicator equal to 1 if the individual declares being white and 0 otherwise, an indicator equal to 1 if at least one of the individual's parents has a doctoral degree and 0 otherwise, an indicator for being an American citizen and 0 otherwise, birth place fixed-effects and doctoral institution fixed-effects.[19] The first-stage equation is commonly estimated with a probit, but I show in Section 7 that results are robust to other specifications.

Figure A.7 shows a histogram of the instrument Z_i which ranges from 0 to 0.93 with a mean of 0.20. Table B.4 shows the first-stage regression. Standard errors are clustered at the PhD major $\times$ PhD cohort level since this is the level of variation of my instrument. A 10 percentage point increase in the instrument is associated with a 2% increase in the likelihood of entering industry, with a Kleibergen-Paap F-statistic equal to 21. Figure A.8 shows histograms of the propensity score, separately for individuals who join industry and individuals who join academia. The two distributions exhibit substantial overlap. The distribution for individuals entering industry is shifted to the right, reflecting their higher likelihood of joining the private sector.[20]

6.2 Alignment Between Academic and Industrial Productivity

Using the estimated propensity score, I estimate earnings within each sector as smooth functions of the propensity score following the method described in Section 3.2. This allows me to recover individuals' predicted earnings in academia and in industry accounting for both observed character-

[19] Birth place is at the state level for individuals who are born in the U.S. and at the country level for individual who are born outside the U.S.

[20] I trim the thinnest tails of support (0.05% of the sample). This is to avoid extrapolation with very few observations.

istics and unobserved selection into sectors, which I use as proxies for sector-specific productivity. The estimation is made using a first-degree polynomial and Section 7 shows that results are robust to other specifications. Figure A.9 presents the distributions of individuals' potential earnings in academia and in industry. Estimations were performed using the log of earnings; the values shown are transformed into thousands of dollars for ease of interpretation. As expected, potential earnings in industry are generally higher than in academia (USD 122,000 on average vs USD 88,000) and have higher variance (standard deviation of USD 47,000 vs USD 25,000).

To what extent does academic productivity reflect a scientist's potential in industry? Because I obtain multiple estimates of potential earnings in each sector for each individual at different points in their career, I first residualize potential earnings in industry and academia by the time varying variables: experience, its square, and PhD cohort fixed-effects. This adjustment removes mechanical life-cycle earnings growth and cohort-specific conditions, yielding an individual-specific measure of earnings potential that is comparable across individuals observed at different career stages. I then average these residualized values across observed periods to obtain a single productivity estimate per sector per individual. This yields a pair of productivity measures for each person, one for academia and one for industry.

I then plot individuals' potential earnings in industry against their potential earnings in academia. Results are shown in Figure 3a. The relationship is positive but weak: the correlation between academic and industrial productivity is 0.29. This magnitude implies substantial uncertainty at the individual level. In particular, an individual with below-median productivity in academia still has a 40% probability of having above-median productivity in industry. I find a similar pattern when academic productivity is proxied using publication output rather than earnings. Using the same MTE approach, I estimate each individual's citation-weighted publication output in academia. To make this measure comparable across individuals, I residualize it by PhD cohort to account for experience and by major, given differences in publication norms across disciplines. This provides a standardized, productivity-based measure of research output in academia over the career. Figure 3b shows the result. The correlation between potential citation-weighted publication output in academia and potential earnings in industry is 0.20. Taken together, these results indicate that academic productivity provides limited information about who will perform well in industry.

To put this magnitude in perspective, I benchmark the academic–industry correlation against predictability within a single sector. Specifically, I examine how predictive early-career earnings are of later earnings within the same sector. After accounting for cohort and life-cycle effects, the correlation between early-career and later-career earnings within the same sector is 0.53, roughly twice as large as the correlation between academic and industrial productivity. Put differently, academic productivity is roughly half as predictive of industrial earnings as early-career earnings are of later earnings within the same sector.

Figure A.13 examines how the correlation differs across fields. It is stronger in Social Sciences (0.52) and in Computer Sciences (0.33) than in Life Sciences (0.19), Physical Sciences (0.17) and Engineering (0.16). Figure 4 shows how this correlation varies across the 56 PhD majors in my dataset.[21] There is substantial variation across majors, with the correlation ranging from -0.04 to 0.58. In some majors, academic productivity is strongly predictive of industrial productivity; in others, it is essentially misleading. The pattern across majors is broadly consistent with differences in the similarity of tasks performed in academia and industry, although, as discussed in Section 2.3, it also depends on how skills co-occur across individuals. Majors such as agricultural economics and industrial psychology exhibit high correlations, consistent with close alignment between academic and industry work. Academic researchers in agricultural economics frequently engage in market analysis, demand and supply forecasting, and policy evaluation related to agricultural and commodity markets, tasks that closely mirror those performed by agricultural economists in industry. Similarly, academic research in industrial psychology focuses on performance measurement, worker selection and incentive design, which map closely to industry roles. In contrast, fields such as computer and systems engineering display a negative correlation. Academic work in these areas tends to emphasize foundational methods and theoretical system design, whereas industry roles are more heavily oriented toward system integration and product development. As a result, academic productivity in these majors can be a poor indicator of productivity in industrial settings.

Finally, I examine how the correlation between academic and industrial productivity has changed over time. I calculate the correlation between residualized potential earnings separately for each PhD cohort. Figure 5 shows the result. This correlation has decreased from 0.40 for cohorts before 1980 to 0.27 for cohorts after 2010, representing a statistically significant 34% decline. This implies that academic productivity has become a less reliable predictor of industrial productivity. This decline is consistent with broader changes in the relationship between academic science and industrial application, including evidence that firms have increasingly shifted away from upstream scientific research toward downstream development and commercialization (Arora et al., 2018).

6.3 Implications for Firms' Hiring Outcomes

In this section, I build on the previous results and examine whether lower alignment between academic and industrial productivity increases the likelihood of hiring individuals who underperform relative to ex-ante expectations. I use layoff incidence as a plausible observable proxy for underperformance, and study how the layoff rate among industry hires varies with the alignment between

[21] 2 majors are dropped because there is no observation to estimate the correlation.

academic and industrial productivity.

A lower alignment between individuals' productivity in academia and in industry implies that academic performance becomes a less reliable guide to industrial productivity. In such settings, firms should be more likely to hire individuals whose academic strengths do not translate into strong performance in industry. To examine this, I use layoffs as a revealed measure of individuals' underperformance. More precisely, each time an individual is surveyed, they report whether they are currently working or not, and if not, the reported reason for unemployment. Using this information, I define a layoff indicator at the person-year level, Layoff_{it}, which equals one if individual i reports not working at time t because of a layoff and 0 if individual i reports working at time t. I interpret a layoff as an observable, realized measure of underperformance, relative to firms' expectations.

I then exploit variation in the alignment between academic and industrial productivity across PhD majors documented in Figure 4 and run the following regression among industry hires:

$$\text{Layoff}_{it} = \alpha + \beta \text{Alignment}_{major(i)} + X_{it} + \varepsilon_{it}$$

$\text{Alignment}_{major(i)}$ is the previously estimated correlation between academic and industrial productivity for major i, computed using estimated earnings in each sector. X_{it} includes: i) PhD cohort and survey year fixed-effects, which capture differences in layoff rate over time and by experience level which could also correlate with the major-level correlation; ii) gender, citizenship, race, parental education and doctoral institution fixed-effects, to account for systematic differences in the composition of graduates across majors that may also correlate with layoffs; iii) location fixed-effects, to account for regional labor-market conditions, since graduates from different majors may sort into different geographic labor markets; iv) PhD field fixed-effects, which absorb systematic differences across fields in layoff risk. As a result, the coefficient β is identified from variation in alignment across majors within the same broad field. I focus on outcomes within the first 15 years of experience, when layoffs are more plausibly linked to initial hiring decisions rather than late-career restructuring or strategic reorganization.[22] Despite the rich set of controls, this analysis remains correlational and is intended to provide suggestive evidence consistent with the proposed mechanism. We expect $\beta < 0$: higher alignment between academic and industrial productivity enables firms to more accurately identify individuals whose skills translate to industry, reducing the likelihood of hiring individuals who subsequently underperform and are laid off.

Column (1) of Table 3 shows the result. I find a negative and statistically significant β coefficient: the higher the alignment between academic and industrial productivity, the lower the layoff rate among industry hires. More precisely, a 0.1 increase in the alignment between academic and

[22]Results are not sensitive to alternative experience cutoffs.

industrial productivity is associated with a 15% reduction in layoff rate relative to the mean. Column (1) of Table B.5 reiterates this analysis using the correlation between individuals' estimated earnings in industry and estimated citation-weighted publication output in academia to calculate the independent variable. Results are similar: a 0.1 increase in the alignment between academic and industrial productivity is associated with a 25% reduction in layoff rate relative to the mean.

As a falsification exercise, I repeat the same analysis using alternative measures of separation that are unlikely to reflect mismatch between workers' productivity and firms' needs. Specifically, I construct indicators for non-employment due to family reasons, illness, or a stated preference not to work. In contrast to layoffs, Tables B.6 and B.7 show no systematic relationship between major-level alignment and these separation outcomes. This pattern supports the interpretation that the negative association between alignment and layoffs reflects demand-side mismatch in firms' hiring decisions – a difficulty in identifying individuals whose productivity translates well to industry – rather than general differences in workers' labor market attachment or exit behavior across majors.

The consequences of variation in the alignment between individuals' productivity in academia and in industry are unlikely to be uniform, as both organizational characteristics and job roles shape how costly mismatch is. I start by examining differences by firm size. I construct an indicator, $\text{Large}_{i,t-1}$, equal to 1 if individual i's last known employer prior to survey wave t employs 25,000 or more employees.[23] I then run the following regression among industry hires:

$$\text{Layoff}_{it} = \alpha + \beta_1 \text{Alignment}_{major(i)} + \beta_2 \text{Large}_{i,t-1} + \beta_3 \text{Alignment}_{major(i)} \times \text{Large}_{i,t-1} + X_{it} + \varepsilon_{it}$$

The marginal effect of alignment on layoff risk is β_1 for small firms and $\beta_1 + \beta_3$ for large firms.

Results are reported in Column (2) of Table 3. Relative to the sample mean, Column (2) implies that a 0.1 increase in alignment is associated with a 41% lower layoff incidence among hires at small firms, whereas the corresponding effect for large firms is statistically indistinguishable from zero. Equivalently, large firms appear insensitive to cross-major variation in alignment. This pattern is consistent with several non-mutually exclusive mechanisms through which large firms may be better able to mitigate the consequences of variation in the informativeness of academic productivity. First, large firms often rely on more formalized, multi-stage screening processes that assess a broader range of job-relevant tasks, reducing their reliance on measures of academic productivity at the point of hire. Second, beyond screening intensity, superior hiring infrastructure, including trained interviewers, accumulated hiring experience, and analytic tools, may allow large firms to more accurately interpret candidate information and evaluate industry-relevant capabilities, limiting the scope for mismatch even when alignment is low. Third, internal labor markets may

[23]For an individual's first observed survey wave, prior employer is unobserved; those observations are dropped.

facilitate ex-post reallocation, reducing the need to layoff when individuals underperform.

I then examine heterogeneity by role, distinguishing between individuals hired in R&D versus non R&D positions after graduation. Results are presented in Column (3) of Table 3 and show a negative and statistically significant relationship between productivity alignment and layoff rates among R&D hires. More precisely, a 0.1 increase in alignment is associated with a 32% lower layoff incidence for R&D hires relative to the mean layoff rate. For non-R&D hires, the effect is statistically indistinguishable from zero. This pattern is consistent with academic productivity being a more central input into screening decisions for R&D roles, where job tasks resemble more closely those performed in academia. In such roles, firms are more likely to rely on academic productivity when hiring individuals, so variation in its informativeness translates into differences in job performance and subsequent layoffs. In non-R&D positions, however, firms are less likely to rely on academic productivity when hiring. As a result, variation in the alignment between academic and industrial productivity has little impact on layoff risk in these roles.

Taken together, these results highlight the challenges firms may face when hiring individuals using performance measures generated in other institutional contexts. In environments where the alignment between academic and industrial productivity is weaker, firms face a higher risk of hiring individuals whose skills do not translate well to industry needs and who subsequently underperform. These challenges seem particularly pronounced for small firms, which have more limited screening capabilities and opportunities for internal redeployment, and in R&D-intensive roles, where firms place greater weight on academic productivity when hiring individuals.

6.4 Signals at Graduation and Industrial Productivity

The previous analyses study the relationship between productivity in academia and productivity in industry, and show that a weaker alignment between academic and industrial productivity is associated with higher layoff rates among industry hires. Building on this result, I now turn to the specific indicators firms can observe at the time of hiring. In particular, I examine the extent to which measures plausibly observable at graduation, such as training history, institutional affiliation, and early research output, are informative about individuals' productivity in industry, and compare this to their informativeness for productivity within academia.

I first regress individuals' potential earnings in industry on PhD cohort, PhD major, demographics, and a set of indicators related to training characteristics (postdoctoral training, doctoral institution's prestige, PhD duration, prior masters'), early research output, funding sources, and preferences over job attributes.[24] Table B.8 shows the results. Several coefficients have signs consistent with

[24]These include: opportunities for advancement, benefits, intellectual challenge, degree of independence, location,

intuition. Postdoctoral training is negatively associated with potential earnings in industry (5% decrease), suggesting that trajectories oriented toward extended academic research are, on average, less aligned with skills valued in industry. By contrast, graduating from a top-20 university is positively associated with potential earnings in industry (+14% increase), indicating that institutional affiliation conveys some information for firms. In contrast, a simple indicator for having at least one publication at graduation is not predictive of potential earnings in industry, highlighting the limited transferability of research-focused academic performance to industrial earnings.

However, the informativeness of such signals for industry productivity is limited. Quantitatively, the set of PhD cohort, PhD major, and demographic controls in the baseline specification already explains a large share of the variation in individuals' estimated earnings in industry (baseline R^2=0.71). Adding the rich set of academic indicators increases explained variation by only a few percentage points, raising R^2 from about 0.71 to 0.78. By contrast, predictive power is notably greater for productivity outcomes within academia. Figure 6 shows that adding the full set of academic indicators increases the R^2 from 0.50 to 0.64 when predicting potential earnings in academia. When predicting potential citation-weighted publication output in academia, the same indicators raise the R^2 from 0.25 to 0.50. Overall, the same academic indicators contribute roughly four times more explanatory power for predicting research output in academia than for predicting earnings potential in industry.

Tables B.9 and B.10 report regressions analogous to Table B.8, using potential earnings and potential citation-weighted publication output in academia as outcomes. Doctoral institution prestige remains highly predictive for both measures. Postdoctoral training is now positively associated with both outcomes (+6% increase). Notably, publication activity at graduation is positively related to potential publication output in academia, even after conditioning on other academic characteristics. This indicates that research-oriented academic signals are informative for predicting future academic impact, while being weak predictors of industry productivity.

7 Robustness

Other measures of productivity: The main results characterize the alignment between productivity in academia and in industry using two proxies for academic productivity: potential earnings and potential publication output. I additionally assess robustness to alternative measures of productivity in industry. First, I estimate individuals' potential citation-weighted publication output in industry. Though publication-based measures are less well suited to capture productivity in industry than

level of responsibility, salary, security and contribution to society. I create an indicator equal to 1 if the individual reports finding the attribute 'very important' and 0 otherwise.

earnings because publishing is not universal and subject to differences in disclosure norms across firms, this provides a useful benchmark for assessing the robustness of the main results. I find a correlation of 0.28 between publication-based productivity measures across the two sectors (Figure A.14), closely aligned with the main findings. In a smaller subsample of 3,599 individuals observed in SDR waves that include patenting information, I estimate individuals' potential patenting output in industry as a proxy for productivity in industry using the same MTE approach. The correlation between potential patenting output in industry and potential publication output in academia equals 0.09 (Figure A.15). The lower correlation in the patent subsample likely reflects differences in field composition, as individuals in the Physical Sciences are slightly over-represented and exhibit lower within-field correlations in my data. Overall, the core patterns are robust to alternative productivity measures and do not hinge on the exclusive use of earnings as a proxy for productivity.

Postdocs: Postdoctoral positions are part of the academic career path and are therefore classified as academic employment in the main analysis. A potential concern, however, is that postdoctoral positions are associated with lower earnings early in the career, which could affect the estimation of earnings potential in academia. Figure A.16 shows that postdoctoral earnings are initially lower than those of individuals who enter academia under a faculty position, but that the two earnings profiles converge within approximately ten years. To assess whether this temporary earnings dip affects the results, I re-estimate the MTE using earnings observations measured after the first ten years post-PhD, thereby focusing on medium-run sectoral earnings rather than early-career wage dynamics. Under this alternative specification, the correlation between potential earnings in academia and industry is 0.28, very similar to the baseline estimate. This indicates that temporary earnings dynamics associated with postdoctoral training do not drive the main results.

Government sector: In my main specification, I bundle individuals who join academia and the government sector together. I also check that my results are robust to excluding individuals who join the government sector (2,930 individuals). I find a similar correlation of 0.31 between potential earnings in academia and in industry. I find a correlation of 0.20 between potential earnings in industry and citation-weighted publications potential in academia, similar to the baseline results.

Functional form assumptions: My baseline Marginal Treatment Effects results use the separate method with a polynomial of degree 1 and a probit for the first-stage equation. Figure A.17 shows that the MTE curves are robust to using a polynomial of degree 2 or a semi-parametric approach in the second-stage, the latter imposing minimal functional form assumptions. I also check in Figure A.18 that the curves are robust to using a logit when estimating the first-stage equation.

Empirical strategy: My instrument exploits variation in individuals' graduation year within a PhD major. A concern is that the composition of skills within a major may evolve over time in ways that correlate with the instrument. For example, a sustained increase in industry demand in a given major could lead more industry-oriented students to sort into that major over time. While my sample ends in 2013, limiting concerns related to more recent shifts such as the rise of artificial intelligence within Computer Science, I directly address this issue by allowing for flexible, field-specific cohort effects. Specifically, I construct indicators for ten-year graduation cohorts (1970–1979, etc.) and interact these cohort indicators with PhD field fixed effects. Including these interactions in the regression allows me to compare individuals within the same field and graduating in the same decade, thereby controlling for slow-moving, field-specific changes in skills composition over time. The F-statistic decreases to 13 but remains strong. The results remain similar, with a correlation of 0.33 between potential earnings in industry and academia, and a correlation of 0.21 between potential citation-weighted publication output in academia and potential earnings in industry.[25]

8 Conclusion

Knowledge workers play an increasingly central role in firm innovation. Yet, when hiring, firms often have limited information about individuals' performance in industry, creating substantial uncertainty when hiring scientific talent from academic environments. What are the consequences of relying on measures of academic productivity when making hiring decisions? This paper investigates the relationship between individuals' productivity in academia and in industry and its consequences for firms' hiring outcomes. Using confidential administrative data on PhD graduates across all fields, linked to wage and publication records, I estimate individuals' productivity in both academia and industry regardless of their realized sector of employment. I do so using a Marginal Treatment Effects framework that accounts for endogenous sorting into sectors and exploits plausibly exogenous variation in relative hiring opportunities between industry and academia at the time of PhD graduation.

I find that individuals' productivity in academia and in industry are weakly aligned, with an average correlation between 0.20 and 0.29. This correlation has decreased over time and varies markedly across majors, ranging from -0.04 to 0.58. The magnitude of the correlation between individuals' productivity in academia and in industry is itself noteworthy. In many settings, performance measures are expected to be noisy across contexts, yet still preserve meaningful information about relative ability. In contrast, the correlations documented here indicate that academic performance provides very little, and sometimes misleading, guidance about how individuals will perform in industry. The results suggest that, even in an environment where firms may observe detailed and standardized indicators of prior performance, translating success across institutional contexts can

[25] Results are robust to shorter time windows, but the instrument becomes weaker.

be far more difficult than commonly assumed. The decline in the correlation between academic and industrial productivity over time also speaks to broader changes in the organization of innovation. My results are consistent with a deeper division of labor between academia and industry, in which the skills rewarded in academia may increasingly differ from those required for productivity inside firms. As a result, academic success may no longer map as closely onto industrial productivity, even as science continues to play an important role in the innovation process (Arora et al., 2018).

Importantly, weaker alignment is associated with a higher incidence of hiring individuals who subsequently underperform, as proxied by layoffs. This pattern is consistent with greater difficulty identifying individuals whose academic strengths translate effectively to industrial work when academic productivity is less informative about industrial productivity. These effects are particularly pronounced when hiring individuals for R&D-intensive occupations and among smaller firms. In these settings, firms are more likely to rely on academic information when making hiring decisions and may have more limited scope for internal reallocation when mismatches arise. Taken together, the results highlight the challenges firms face when hiring scientific talent using performance indicators generated in a different institutional context.

My results underscore the challenges firms face when hiring individuals from a different institutional context. When available measures of productivity are shaped by norms and incentives that differ from those governing performance inside firms, they can increase the risk of hiring mismatches. Importantly, the difficulty does not arise because academic productivity is an imprecise signal of a single underlying ability. Rather, it reflects the fact that academic and industrial productivity capture different dimensions of performance shaped by distinct institutional norms and incentives. As a result, even perfectly measured academic outcomes may provide limited guidance about who will perform well in industrial settings. This tension arises in many settings where organizations must assess individuals across institutional boundaries, such as when firms draw on scientific research, external expertise, or knowledge developed outside their own organizational context (Bikard, 2018; Bikard and Marx, 2020; Boudou and Roche, 2023; Cooper et al., 1994; Gittelman and Kogut, 2003). By documenting how productivity aligns across academia and industry, the paper clarifies a fundamental constraint on firms' ability to leverage external sources of talent.

Methodologically, the paper uses a Marginal Treatment Effects framework to recover individuals' productivity in both academia and industry, independent of their realized sector of employment. This approach allows me to move beyond average effects and to capture individual-level heterogeneity. More broadly, the findings highlight that leveraging scientific talent is not simply a matter of acquiring highly trained individuals. Rather, it depends critically on firms' ability to identify those scientists whose skills align with the firm's specific institutional environment and strategic needs (Arora and Gambardella, 1994). In this sense, the paper points to the evaluation of individuals across institutional contexts as a core strategic capability in innovation-driven organizations.

These findings also have significant implications for firms' hiring strategies, especially as organizations increasingly rely on knowledge and innovation to drive performance. When measures of performance originate from a different institutional context, even highly structured hiring processes can lead to suboptimal outcomes - including costly layoffs and missed opportunities to recruit high-potential talent. These challenges may be further amplified in the current hiring environment, where the growing use of generative AI has made traditional evaluation tools (e.g., take-home coding tests or written assessments) less reliable for identifying high-potential candidates. As a result, firms may default to readily observable academic credentials (Piezunka and Dahlander, 2015), such as publications or university prestige, that may not always translate into industrial performance. My findings underscore the risks of this approach and highlight the importance of tailored evaluation practices. The variation in alignment between academic and industrial productivity across majors suggests that firms may benefit from major-specific hiring strategies to assess candidates. These may include interviews that simulate real work constraints, internships, or collaborative research projects that allow firms to directly observe candidates' performance in relevant contexts (Baruffaldi and Poege, 2025; Honoré and Ganco, 2023).

A few limitations of this study provide fertile ground for future work. First, given constraints on the data used, I do not have information about organization-level outcomes, which limits the ability to concretely link scientists' sorting to firm performance. In particular, the study does not address whether the current allocation of talent across sectors is optimal, either from the perspective of firms or from that of social planners. Moreover, I lack information about the characteristics of individuals' publication output, such as journals, references, and abstracts. This limitation restricts my ability to discuss the direction of the innovation that individuals would produce within each sector. Assessing these directional aspects of innovation and how they relate to the types of talent they recruit might be crucial for organizations, especially in firms where scientists have autonomy over the direction of their research. These gaps highlight promising avenues for future research.

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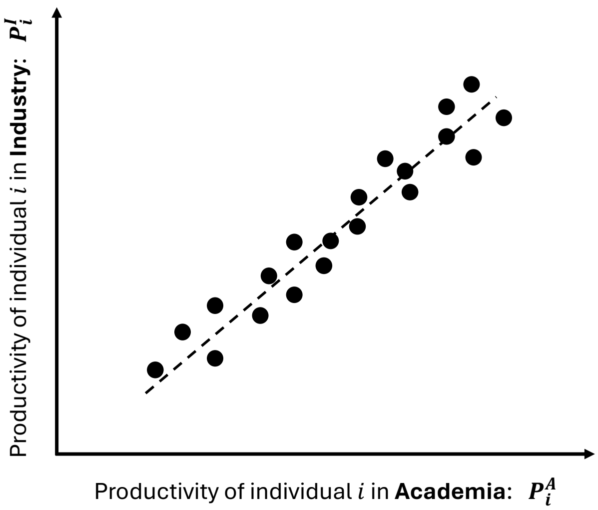
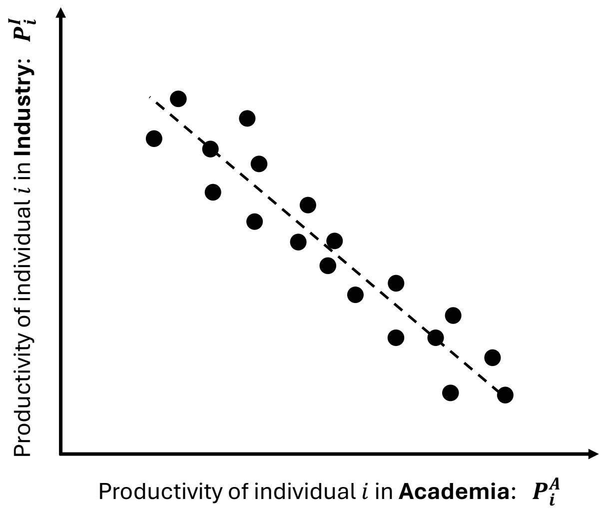
Figures and Tables

Figure 1: Stylized relationship between individuals' productivity in academia and in industry

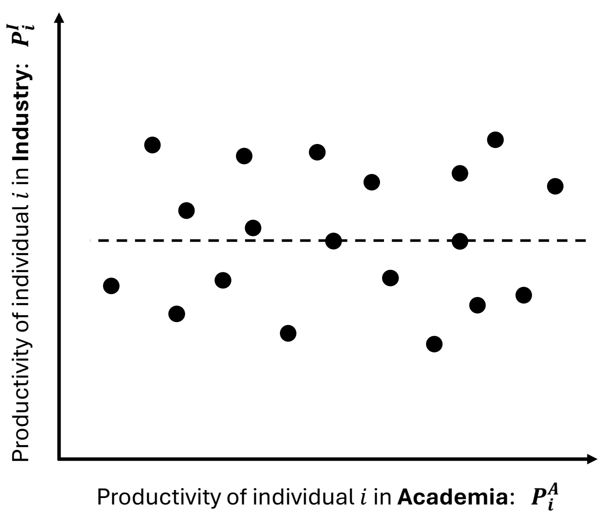
(a) Positive correlation



(b) Negative correlation

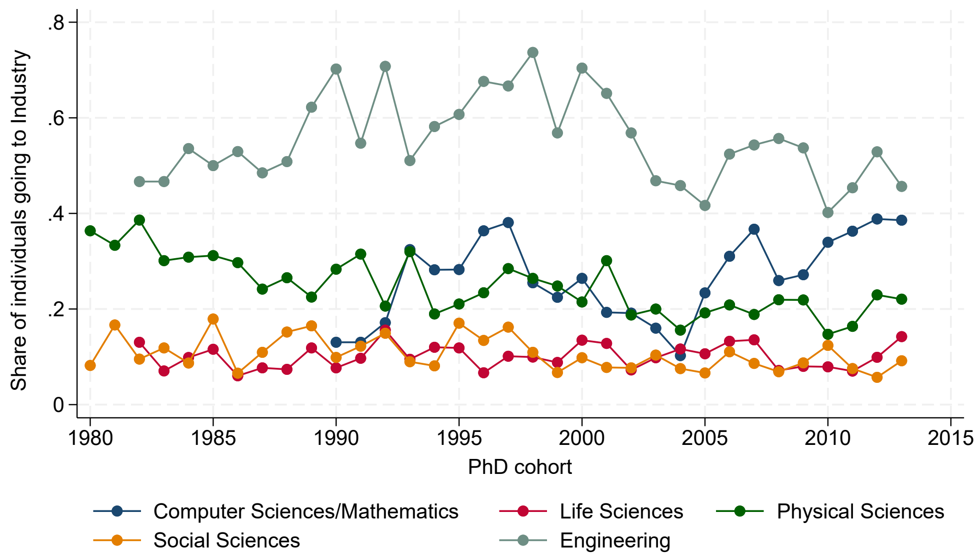


(c) Null correlation



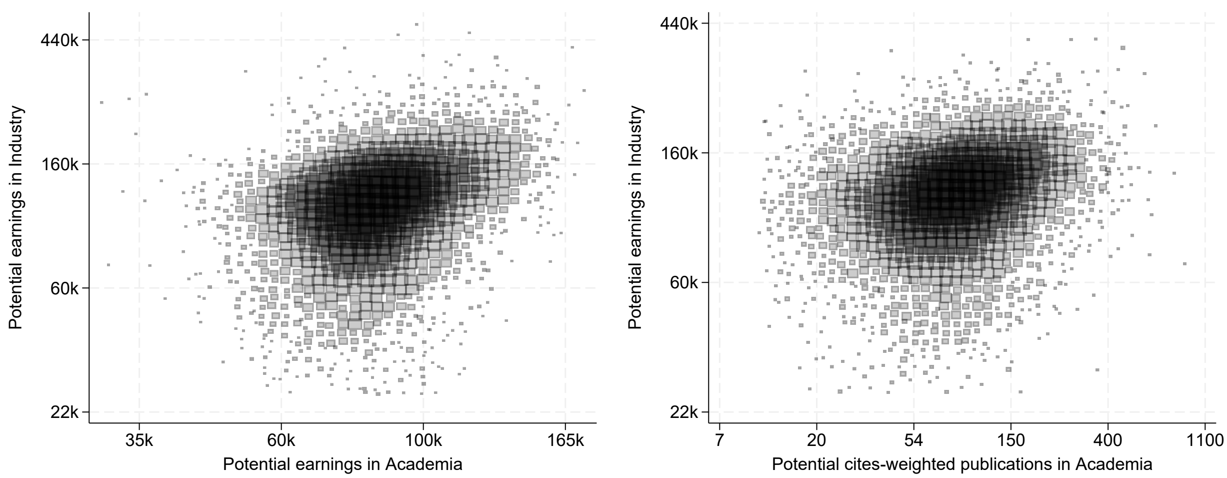
Notes: Each dot represents an individual. Each panel illustrates a stylized joint distribution between individuals' productivity in academia and individuals' productivity in industry. (a) Positive correlation: individuals productive in one sector are also productive in the other. (b) Negative correlation: individuals tend to be productive in one sector but not the other. (c) Null correlation: no systematic relationship between individuals' productivity across sectors.

Figure 2: Share of individuals joining industry, by PhD cohort and field



Notes: This figure shows the share of individuals joining industry by PhD graduation cohort and PhD field. Dots are omitted for early cohorts in certain fields due to confidentiality restrictions.

Figure 3: Relationship between individuals' productivity in academia and in industry

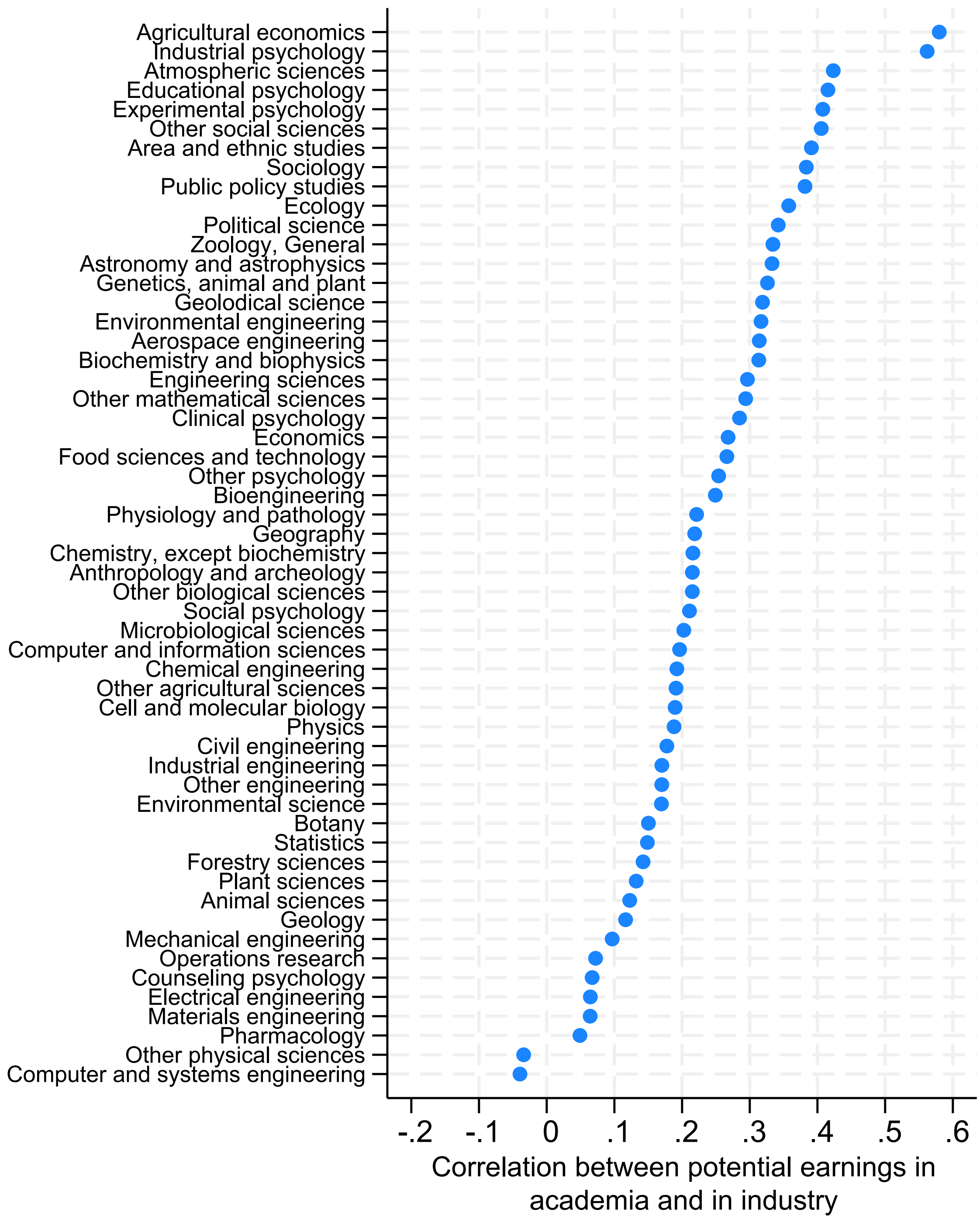


(a) Using potential earnings

(b) Using potential earnings and publications

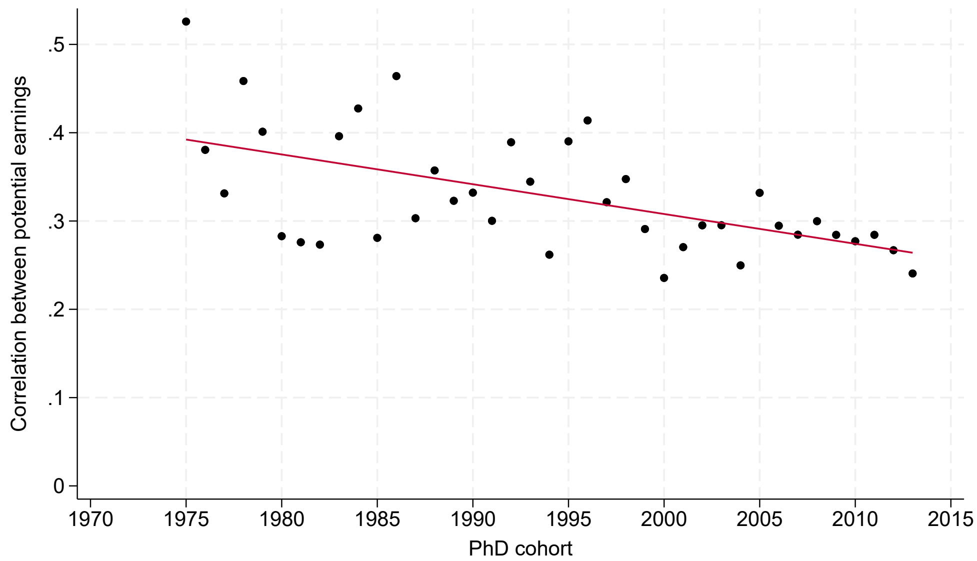
I

Figure 4: Correlation between individuals' productivity in academia and in industry, by PhD major



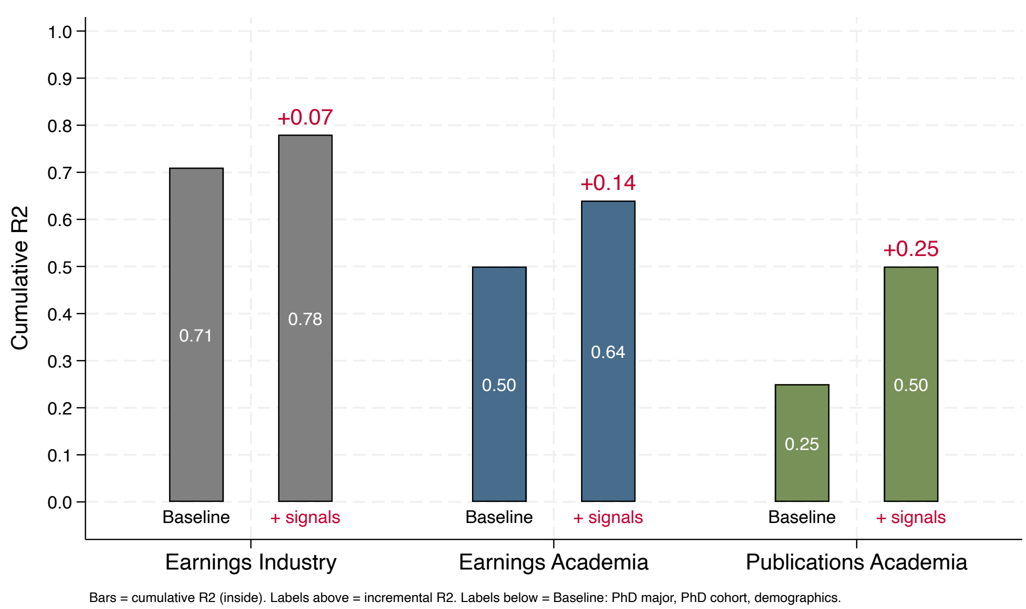
Notes: This figure reports the correlation between individuals' productivity in academia and in industry, as a function of PhD major. Individuals' productivity in academia and industry is proxied by their potential earnings in each sector, residualized by PhD cohort, experience and experience squared, and averaged at the individual level.

Figure 5: Correlation between individuals' productivity in academia and in industry, by PhD cohort



Notes: This figure reports the correlation between individuals' productivity in academia and in industry, by PhD cohort of graduation. Individuals' productivity in academia and industry is proxied by their potential earnings in each sector, residualized by PhD cohort, experience and experience squared, and averaged at the individual level.

Figure 6: Informational content of academic signals for predicting sector-specific outcomes



Notes: Bars report cumulative R^2 from regressions of individuals' potential outcomes on observable characteristics. Baseline controls include PhD cohort, PhD major and demographics (gender, race, citizenship, parental education). "+ Signals" adds academic indicators observable at PhD graduation, including training history, early publications, funding sources, and job preferences. Numbers above bars indicate the incremental contribution of these signals to explained variation. See Section 6.4 for more details.

Table 1: Summary statistics

	(1) Academia		(2) Industry		(1)-(2) Difference	
	mean	sd	mean	sd	diff in means	t-stat
Earnings (th., 2015 USD)	99.9	58.2	167.4	102.6	-67.48	-43.4***
Publications (cum.)	20.8	30.4	5.3	11.2	15.50	49.3***
PhD Graduation Year	2000.4	10.5	2000.6	9.9	-0.11	-0.7
White	0.7	0.5	0.6	0.5	0.11	13.3***
Female	0.4	0.5	0.3	0.4	0.18	23.6***
Mother has PhD	0.1	0.2	0.1	0.2	0.01	2.8**
Father has PhD	0.2	0.4	0.2	0.4	0.03	5.0***
American	0.8	0.4	0.6	0.5	0.13	16.4***
Observations	17,895		4,714		22,609	

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. This table lists summary statistics for the sample of 22,609 individuals, separately for individuals joining academia and individuals joining industry. Column 'diff in means' reports the difference between the mean in academia and the mean in industry. Column 't-stat' reports the t-statistic of a test of equality of means. *Earnings*: Average annual earnings in thousands of USD 2015, computed across all survey waves in which the individual is observed. *Publications*: aggregate number of publications between PhD graduation and 2017. *PhD graduation year*: Fiscal year of doctorate. *White*: 0/1=1 if the individual's race/ethnicity is reported as white. *Female*: 0/1=1 if the individual reports being a female. *Mother has PhD*: 0/1=1 if the individual's mother highest educational attainment is a PhD. *Father has PhD*: 0/1=1 if the individual's father highest educational attainment is a PhD. *American*: 0/1=1 if the individual is a U.S. citizen.

Table 2: Falsification tests - Independence assumption

Test of:	Strategic Graduation			Strategic Entry	
	PhD Duration	$<$ Average	$>$ Average	Z_i	Industry
	(1)	(2)	(3)	(4)	(5)
Z_i	-0.068	0.067	-0.057		
	(0.263)	(0.069)	(0.070)		
Z_i (entry)				-0.018	-0.003
				(0.020)	(0.042)
PhD major FE	Yes	Yes	Yes	Yes	Yes
PhD cohort FE	Yes	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes	Yes
Doctoral institution FE	Yes	Yes	Yes	Yes	Yes
Observations	16,733	16,720	16,720	14,279	14,279

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The outcome in Column (1) is calculated as the difference between PhD graduation year and PhD entry year. The outcome in Column (2) is an indicator equal to 1 if PhD duration is strictly below the average PhD duration of other individuals in the same cohort and PhD major. The outcome in Column (3) is an indicator equal to 1 if PhD duration is strictly above the average PhD duration of other individuals in the same cohort and PhD major. The outcome in Column (4) is the instrument, calculated as in Equation 1. The outcome in Column (5) is an indicator equal to 1 if the individual joins industry. Z_i *(entry)* is the instrument calculated for the year before PhD entry. *Demographics* controls for race, gender, parental education, citizenship and birth place. All regressions also control for field-specific federal and private R&D funding at the time of graduation. Standard errors are clustered at the PhD major$\times$PhD cohort level.

Table 3: A weaker alignment between individuals' productivity in academia and in industry is associated with higher layoff rates, especially for small firms and R&D occupations

	Layoff$_{it}$ $= 1$		
	(1)	(2)	(3)
Alignment$_{major(i)}$	-0.015*	-0.041**	0.008
	(0.009)	(0.018)	(0.017)
Large Firm=1		-0.011***	
		(0.004)	
Alignment$_{major(i)}$ $\times$ Large Firm=1		0.056**	
		(0.024)	
R&D occupation=1			0.004
			(0.003)
Alignment$_{major(i)}$ $\times$ R&D occupation=1			-0.032**
			(0.015)
PhD cohort FE	Yes	Yes	Yes
Survey year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
PhD field FE	Yes	Yes	Yes
Observations	10,274	6,611	9,979
Mean of dep. var.	0.01	0.01	0.01

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The dependent variable is an indicator equal to 1 if individual i reports not working in year t because of a layoff and 0 if individual i reports working. The independent variable is the correlation between individuals' productivity in academia and in industry, calculated at the PhD major level, showed in Figure 4. Individuals' productivity in each sector is proxied by their potential earnings in each sector, residualized by PhD cohort, experience and experience squared, and averaged at the individual level. Higher correlation reflects stronger alignment. Standard errors are clustered at the PhD major level.

Not All That Glitters is Gold:
Firm Hiring in the Market for Knowledge Workers

Appendix

Appendix A Figures

Figure A.1: PhD fields and majors

Field	Major
(shaded box)	(white box)

Computer Sciences/Mathematics
Computer/Information Sciences
Applied Mathematics
Operations Research
Statistics
Other Mathematics

Life Sciences
Biochemistry and Biophysics
Biology
Botany
Cell and Molecular Biology
Ecology
Genetics
Microbiological Sciences
Animal Sciences
Food Sciences
Plant Sciences
Other Agricultural Sciences
Pharmacology
Zoology
Other biological Sciences
Physiology and Pathology
Environmental Sciences
Forestry Sciences

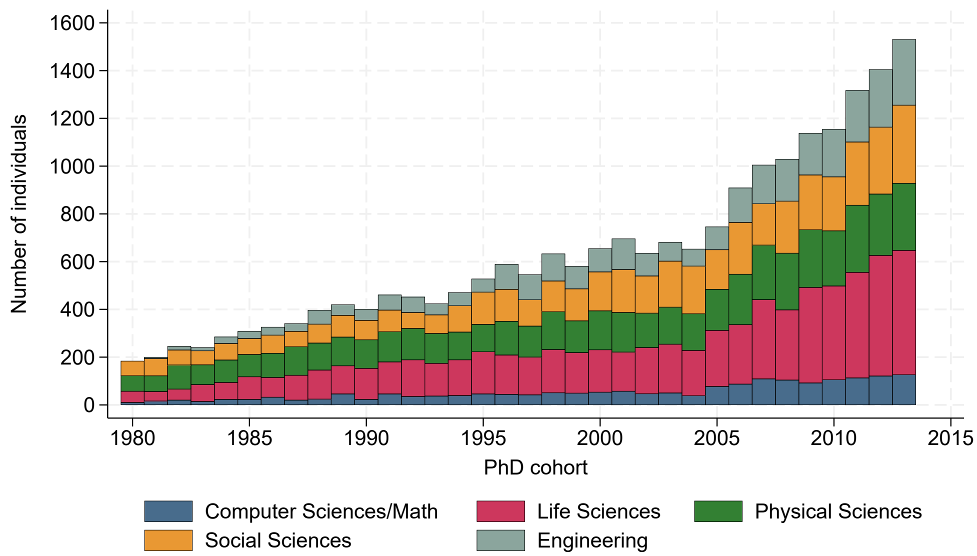
Physical Sciences
Astronomy/Astrophysics
Physics
Other physical sciences
Chemistry
Atmospheric Sciences
Geology
Geological Sciences

Engineering
Aerospace Engineering
Chemical Engineering
Civil Engineering
Computer Engineering
Electrical Engineering
Industrial Engineering
Mechanical Engineering
Bioengineering and biomedical Engineering
Engineering Sciences
Environmental Engineering
Materials Engineering
Other Engineering

Social Sciences
Agricultural economics
Economics
Public Policy
Political Sciences
Educational Psychology
Clinical Psychology
Counseling psychology
Experimental psychology
Industrial/Organizational psychology
Social psychology
Other psychology
Anthropology/Archaeology
Criminology
Sociology
Area and Ethnic Studies
Geography
Other social sciences

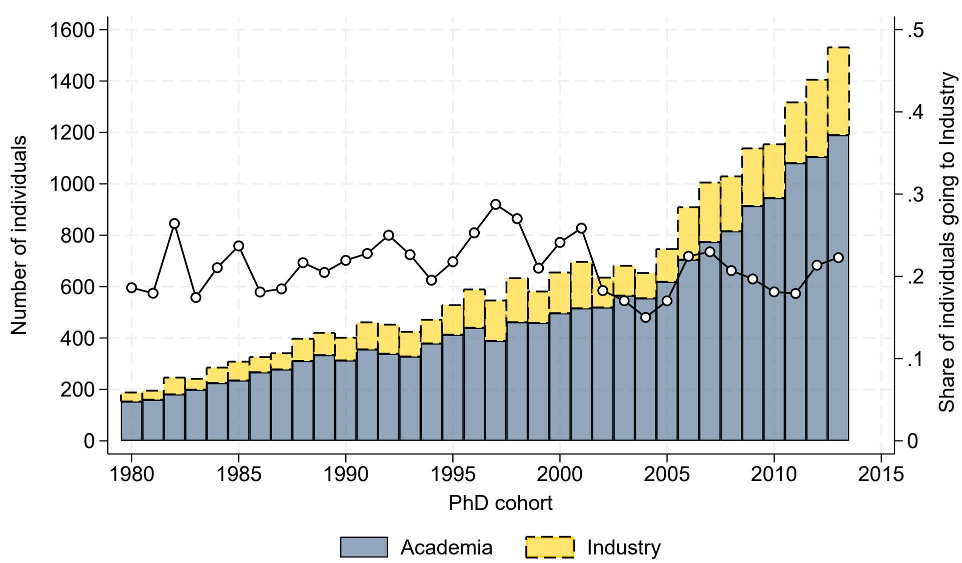
Notes: This figure shows the PhD fields and majors present in my sample. For confidentiality reasons, I am not allowed to export the number of students per PhD major, but Figure A.2 shows the number of individuals by PhD cohort of graduation and main PhD field of study.

Figure A.2: Number of individuals, by PhD cohort and field



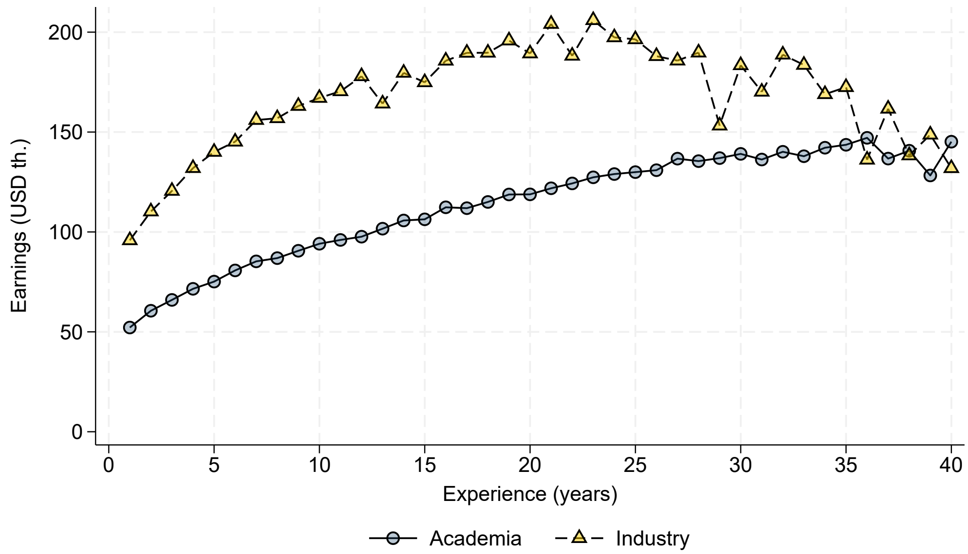
Notes: This figure shows the number of individuals by PhD cohort of graduation and main PhD field of study. For confidentiality reasons, the values corresponding to the years 1980 and 1981 for Engineering are combined into a single dot showed in 1981.

Figure A.3: Sector joined, by PhD cohort



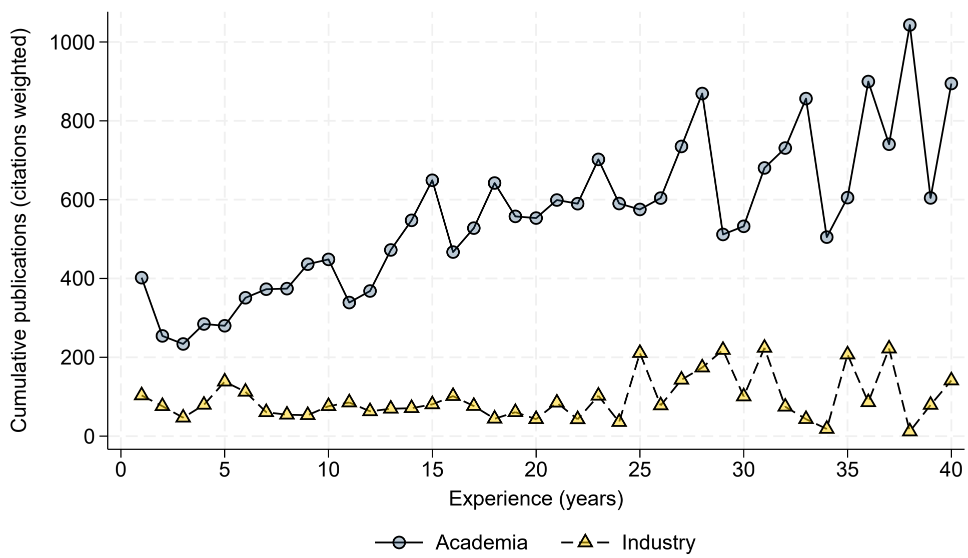
Notes: This figure shows the number of individuals entering academia (blue bars) and industry (yellow bars) by PhD graduation cohort. The black line shows the share of individuals entering industry. Academia and industry are defined in Section 4.3.

Figure A.4: Earnings as a function of experience and sector joined at graduation



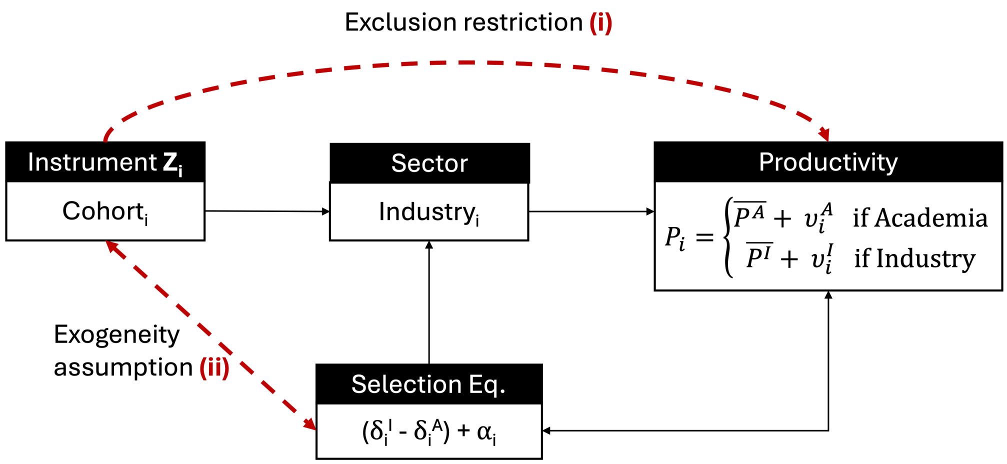
Notes: This figure shows average earnings in thousands of USD 2015 as a function of sector joined and experience.

Figure A.5: Citation-weighted publications (stock) as a function of experience and sector joined at graduation



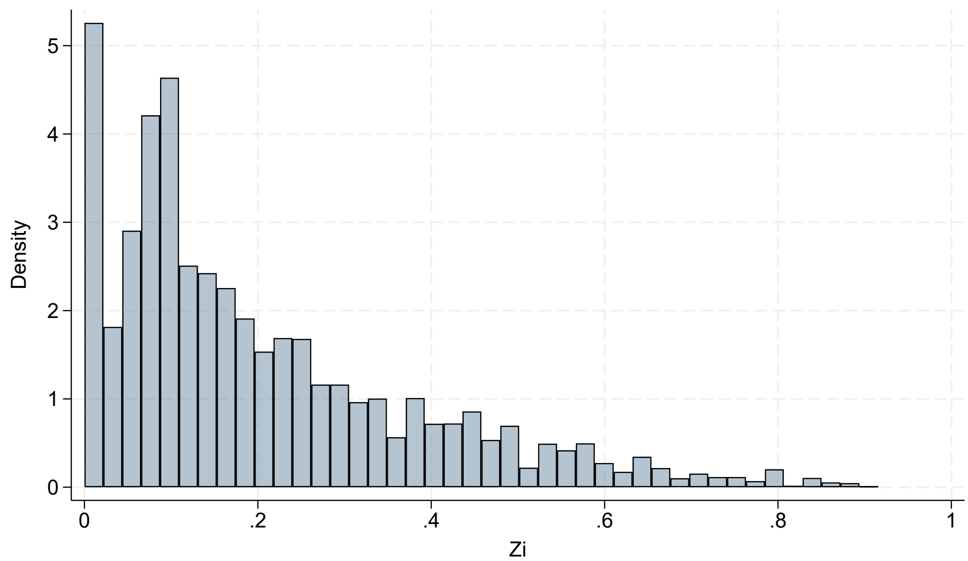
Notes: This figure shows the cumulative number of publications published by individuals between PhD graduation year and 2017, as a function of experience and sector joined. The number of publications is weighted by the average number of citations received up to 5 years after publication.

Figure A.6: Instrumental variable, exogeneity assumption



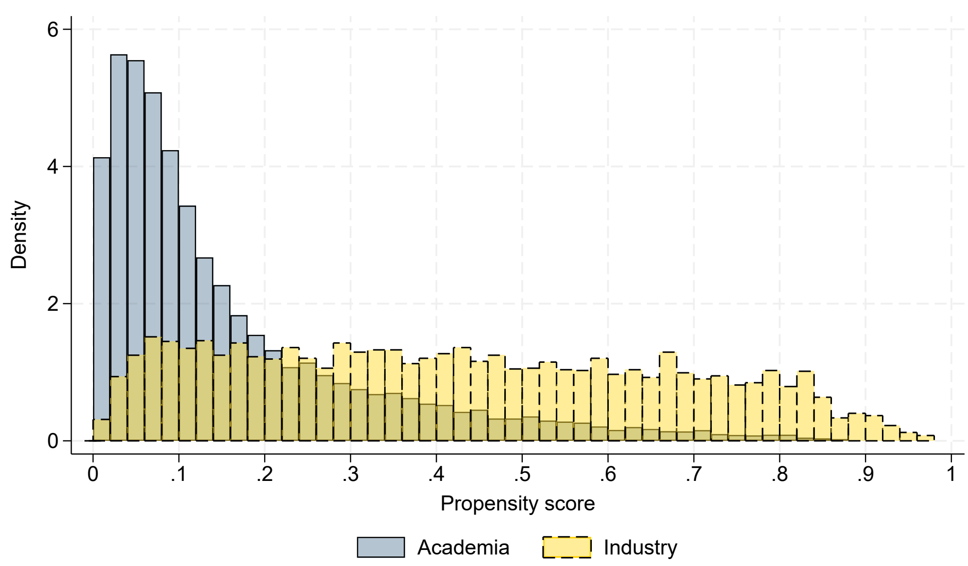
Notes: The utility function of individual i in sector j is $U_i^j = \beta_i W_i^j + \alpha_i 1\{j = \text{industry}\}$, where α_i captures individual i's preference for industry relative to academia and β_i reflects sensitivity to earnings. W_i^j can be decomposed into $W_i^j = \overline{W^j} + \delta_i^j$ where $\overline{W^j}$ captures the part of earnings explained by observables and δ_i^j captures the part of earnings related to unobservable components. Similarly, the productivity of individual i in sector j can be written as $P_i^j = \overline{P^j} + v_i^j$. Individual i joins industry iff $\beta_i(\overline{W^I} - \overline{W^A}) + \beta_i(\delta_i^I - \delta_i^A) + \alpha_i > 0$. Section D shows that $\delta_i^I - \delta_i^A$ is correlated with δ_i^I and/or δ_i^A, which creates endogeneity issues when productivity is proxied by earnings, i.e., $P_i^j = W_i^j$. As a consequence, the exogeneity assumption requires that individuals who select into industry (higher $\delta_i^I - \delta_i^A$) do not systematically graduate in years where industry hiring opportunities are high.

Figure A.7: Distribution of the instrumental variable Z_i



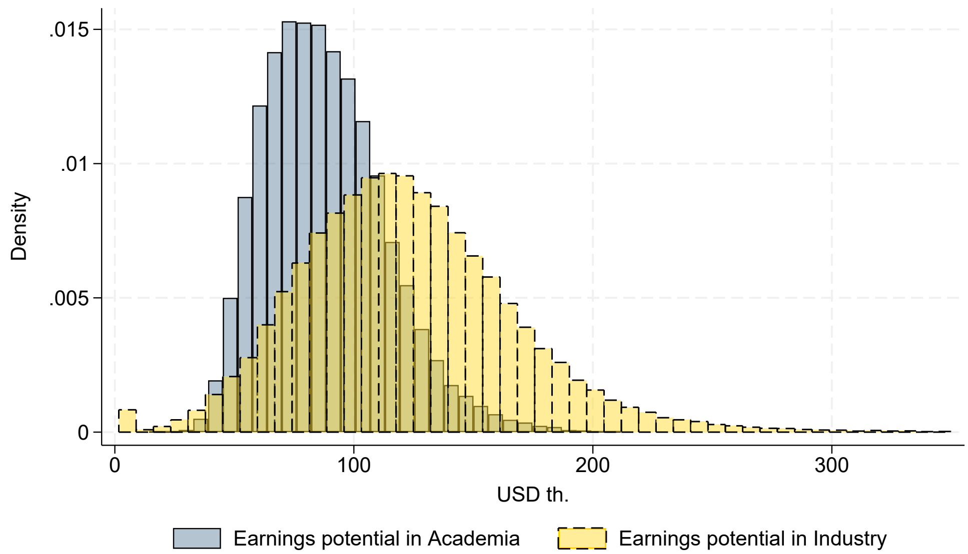
Notes: This figure shows an histogram of the instrumental variable Z_i as defined in Equation 1 for the 22,609 unique individuals in the sample. Bars with less than 5 observations are excluded for confidentiality reasons.

Figure A.8: Distribution of estimated propensity to join industry, by sector joined



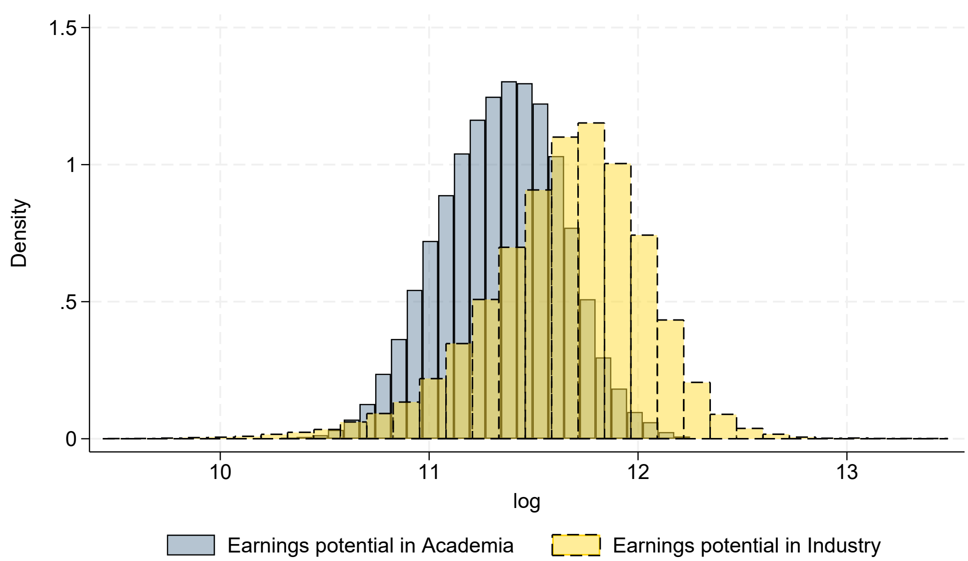
Notes: This figure displays the distribution of the estimated propensity score - the predicted probability that an individual enters industry at graduation - based on observables (X_i) and the instrument (Z_i). The distribution is plotted separately for individuals who join academia (blue bars with solid lines) and those who join industry (yellow bars with dashed lines). For confidentiality reasons, only bars with at least 5 observations are displayed.

Figure A.9: Distribution of potential earnings in academia and industry, in USD



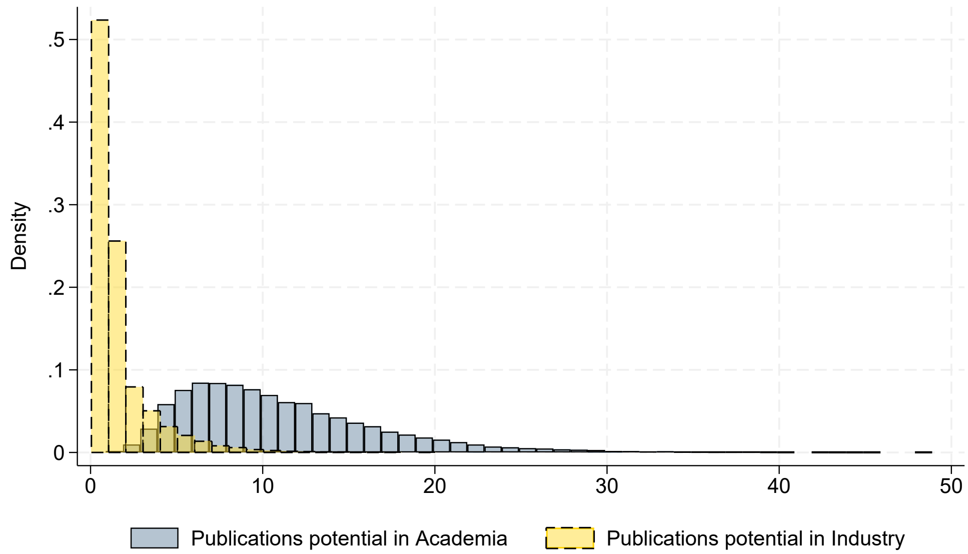
Notes: This figure shows the distribution of individuals' potential earnings in academia, transformed in 2015 USD thousands (blue bars, solid line) and potential earnings in industry, transformed in 2015 USD thousands (yellow bars, dashed line).

Figure A.10: Distribution of potential earnings (log) in academia and industry



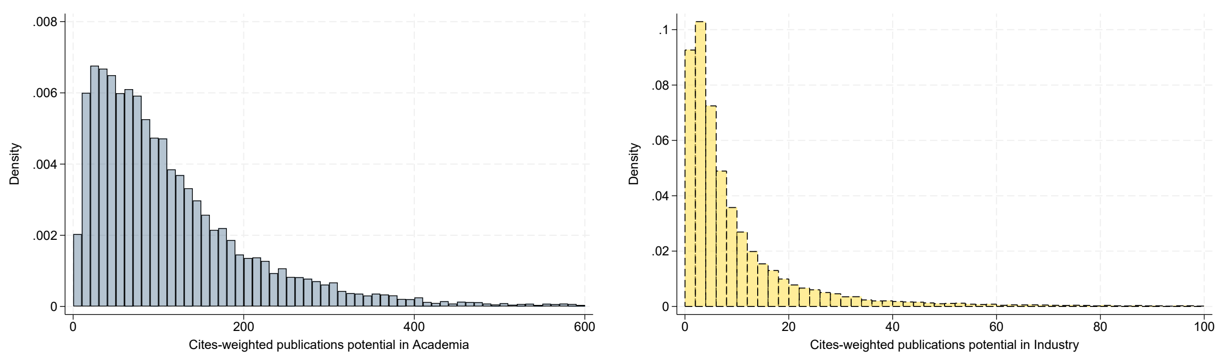
Notes: This figure shows the distribution of potential earnings in academia, in log (blue bars, solid line) and potential earnings in industry, in log (yellow bars, dashed line).

Figure A.11: Distribution of potential publications in academia and industry



Notes: This figure shows the distribution of potential publication output (from graduation until 2017) in academia (blue bars, solid line) and potential publication output (from graduation until 2017) in industry (yellow bars, dashed line).

Figure A.12: Distribution of potential citation-weighted publication output in academia and industry



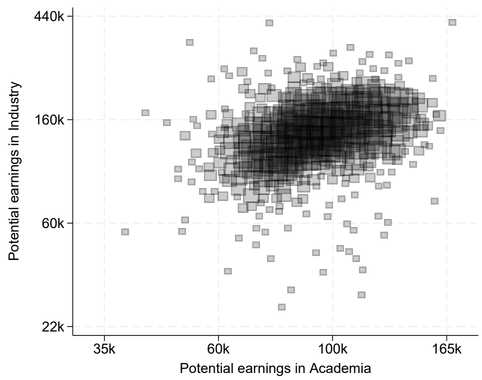
(a) Academia

(b) Industry

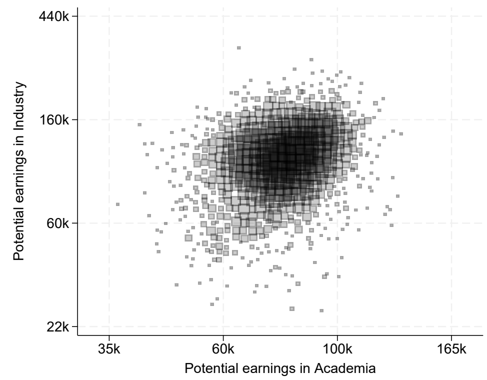
Notes: This figure shows the distribution of potential citation-weighted publication output (from graduation until 2017) in academia (blue bars, solid line) and potential citation-weighted publication output (from graduation until 2017) in industry (yellow bars, dashed line).

Figure A.13: Relationship between individuals' productivity in academia and industry, by field

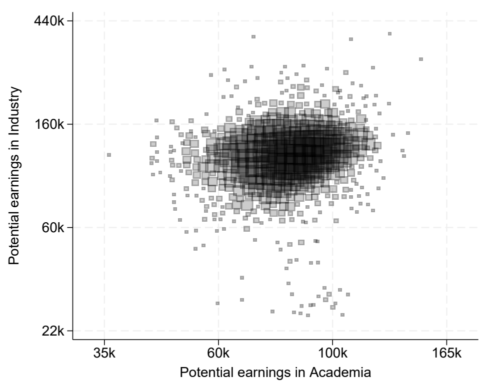
(a) Computer and Mathematical Sciences



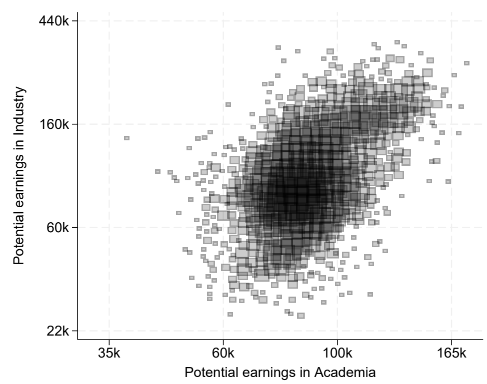
(b) Life Sciences



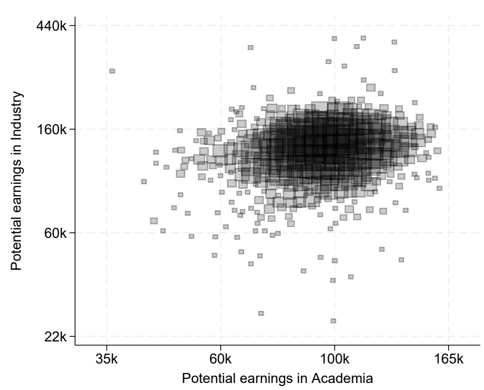
(c) Physical Sciences



(d) Social Sciences

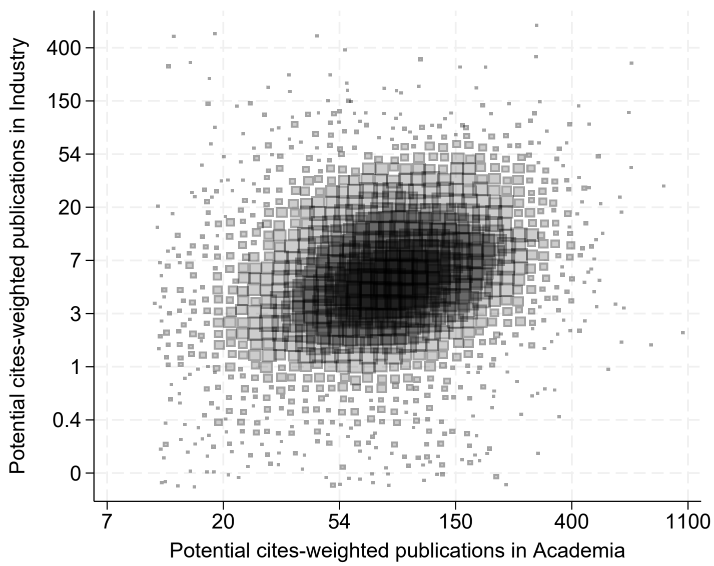


(e) Engineering



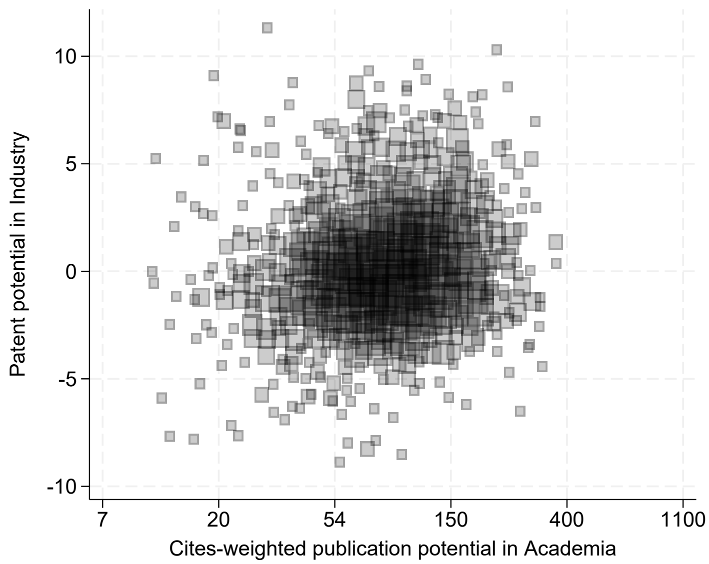
Notes: This figure shows a scatter plot of individuals' productivity in academia (x-axis) and industry (y-axis), by PhD field. Productivity in each sector is proxied by potential earnings in that sector, residualized by PhD cohort, experience and experience squared, and averaged at the individual level. I add back the mean earnings in each sector to make the scales easier to interpret after residualizing.

Figure A.14: Relationship between individuals' productivity in academia and in industry, using potential citation-weighted publication output in each sector to proxy for productivity



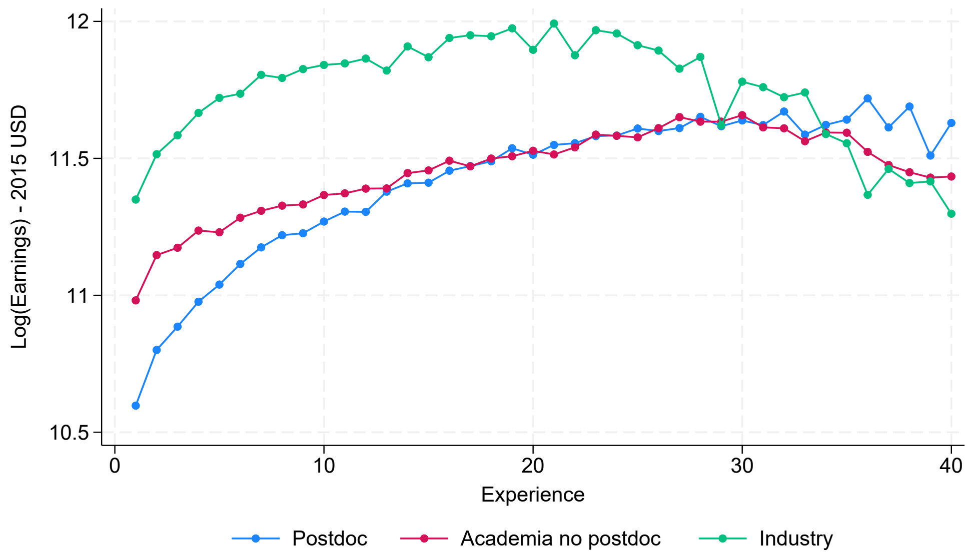
Notes: This figure shows a scatter plot of individuals' productivity in academia (x-axis) and industry (y-axis). Productivity in each sector is proxied by potential publication output in that sector from graduation to 2017, weighted by citation counts received within five years and residualized by PhD cohort and PhD major. I add back the mean citation-weighted publications in each sector to make the scales easier to interpret after residualizing.

Figure A.15: Relationship between individuals' productivity in academia and in industry, using potential citation-weighted publication output in academia and potential patent output in industry



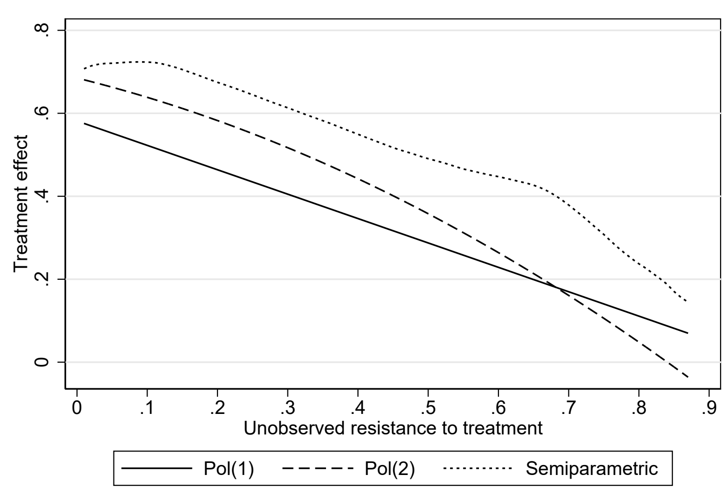
Notes: This figure shows a scatter plot of individuals' productivity in academia (x-axis) and industry (y-axis). Productivity in industry is proxied by potential patent applications in industry, residualized by PhD cohort, experience, its square and PhD major, and averaged at the individual level. Productivity in academia is proxied by potential publication output in academia from graduation to 2017, weighted by citation counts received within five years and residualized by PhD cohort and PhD major. I add back the mean citation-weighted publications in academia and the mean number of patent applications in industry to make the scales easier to interpret after residualizing.

Figure A.16: Earnings curve by employment status upon PhD graduation



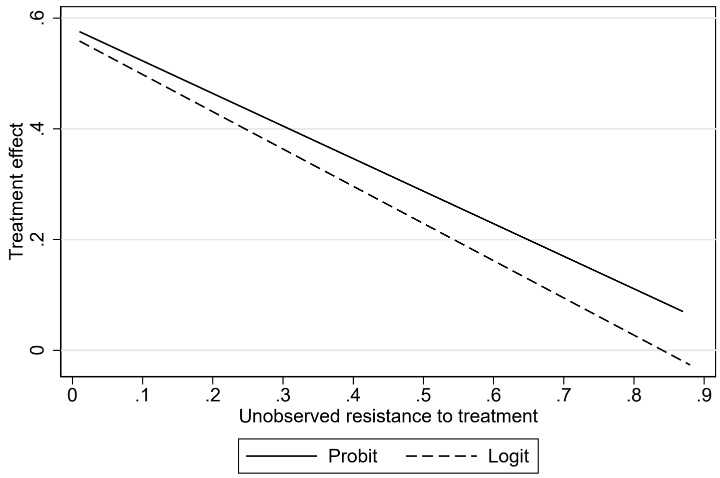
Notes: This figure shows average earnings of individuals as a function of their employment status upon PhD graduation and experience. The blue curve represents individuals who start a postdoc after PhD graduation. The red curve represents individuals who join academia after PhD graduation but not under a postdoc position. The green curve represents individuals who join the private sector. Earnings are in 2015 USD, winsorized below the 1^{st} and above the 99^{th} percentiles and logged. Experience is defined as survey year minus PhD cohort.

Figure A.17: Marginal Treatment Effects curve using different functional forms in the second-stage, Log(Earnings)



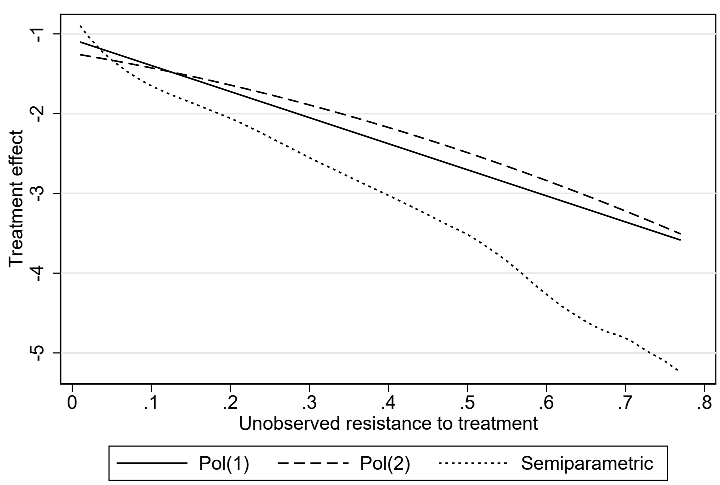
Notes: This figure shows the MTE curve using a polynomial of degree 1 (solid line), a polynomial of degree 2 (dashed line) and a semiparametric approach (dotted line). The dependent variable of interest is Log(Earnings).

Figure A.18: Marginal Treatment Effects curve using different first-stage specifications, Log(Earnings)



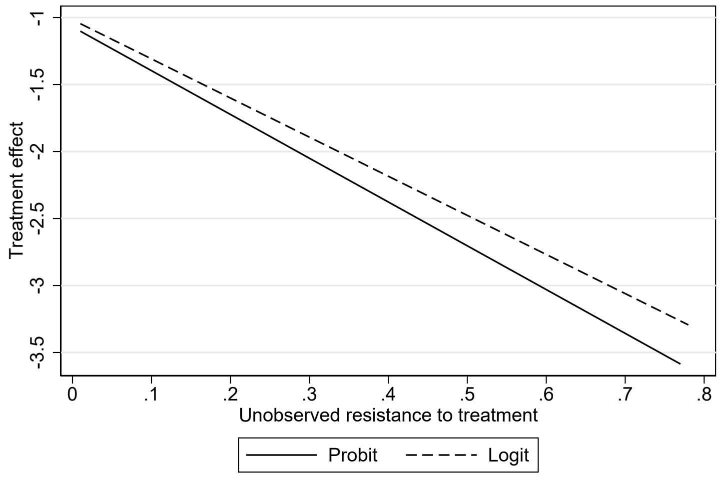
Notes: This figure shows the MTE curve using a probit function (solid line) and a logit function (dashed line) for the specification of the first-stage. The dependent variable of interest is Log(Earnings).

Figure A.19: Marginal Treatment Effects curve using different functional forms in the second-stage, Log(citation-weighted publications)



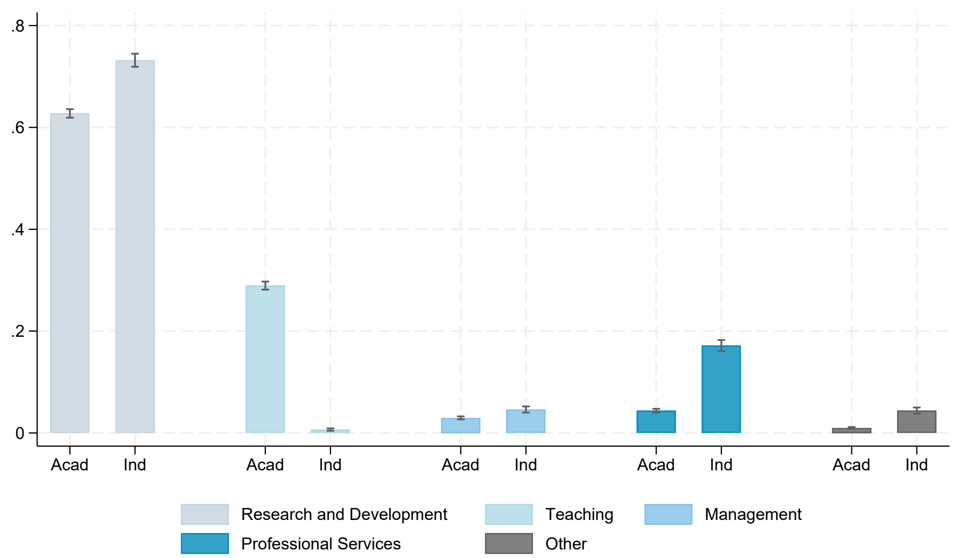
Notes: This figure shows the MTE curve using a polynomial of degree 1 (solid line), a polynomial of degree 2 (dashed line) and a semiparametric approach (dotted line). The dependent variable of interest is Log(Citation-weighted publication output).

Figure A.20: Marginal Treatment Effects curve using different first-stage specifications, Log(citation-weighted publications)



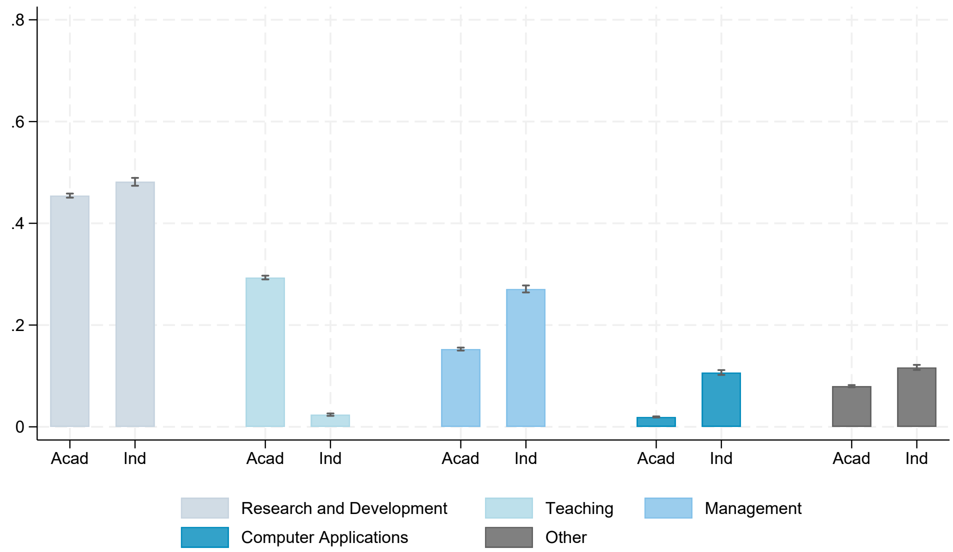
Notes: This figure shows the MTE curve using a probit function (solid line) and a logit function (dashed line) for the specification of the first-stage. The dependent variable of interest is Log(Citation-weighted publication output).

Figure A.21: Main activity (first employment), by sector joined



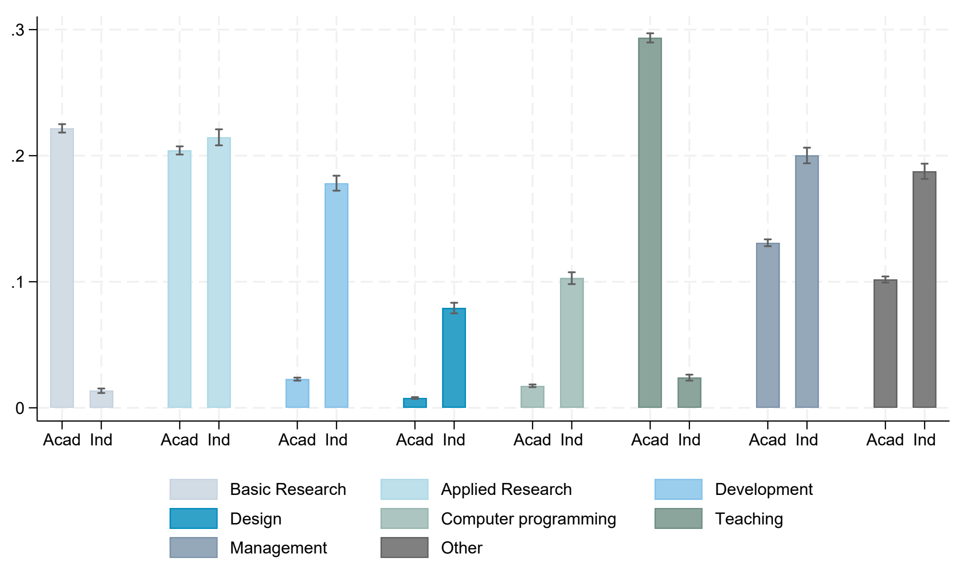
Notes: This figure shows individuals' main activity as a function of sector joined at graduation upon PhD graduation. *Acad* represents academia and *Ind* represents industry.

Figure A.22: Main activity (at the time of survey), by sector joined



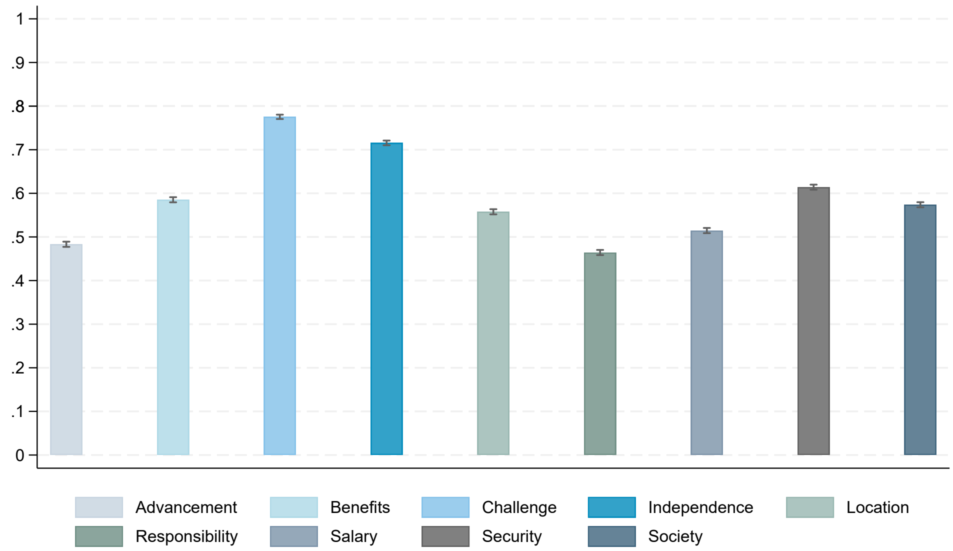
Notes: This figure shows individuals' main activity when surveyed during the career. *Acad* represents academia and *Ind* represents industry. Averages are taken across all observations I have of individuals.

Figure A.23: Detailed activity (at the time of survey), by sector joined



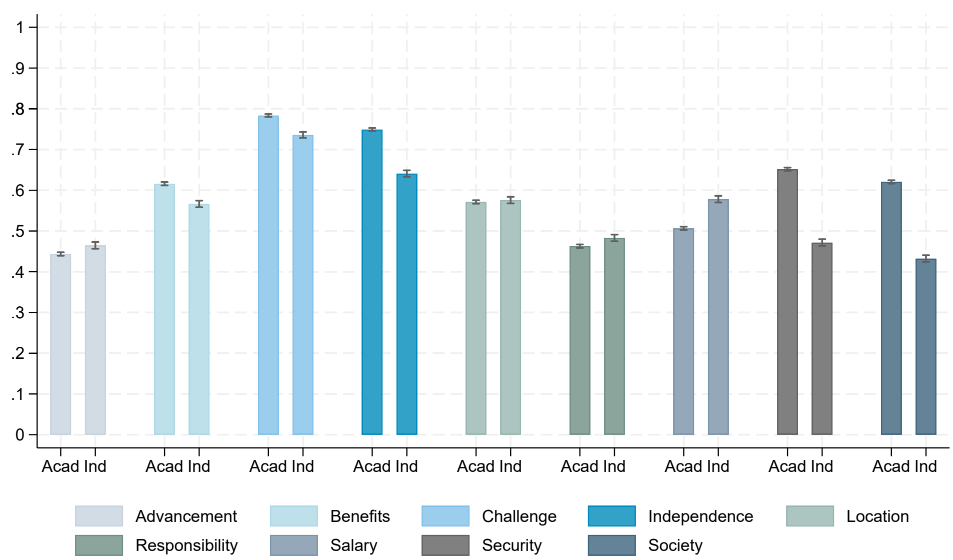
Notes: This figure shows individuals' work activity in which they spent most hours on when surveyed during the career. *Acad* represents academia and *Ind* represents industry. Averages are taken across all observations I have of individuals.

Figure A.24: Preferences over job attributes



Notes: This figure shows the share of individuals who report the job attribute as 'very important'. Possible answers are 'very important', 'somewhat important', 'somewhat unimportant' or 'not important at all'.

Figure A.25: Preferences over job attributes, by sector joined



Notes: This figure shows the share of individuals who report the job attribute as being 'very Important', as a function of sector joined. Averages are taken across all observations I have of individuals.

Appendix B Tables

Table B.1: Sample Construction

Sample restriction	Individuals excluded	% of initial sample
Exclude Health-related PhD majors	3,626	4.6%
Exclude cohorts graduating before 1970	1,277	1.6%
No definite post-graduation commitment	24,621	31.4%
Exclude internships, residencies, traineeships, or military service	2,527	3.2%
Missing employment information	2,364	3.0%
Exclude individuals starting careers outside the U.S.	5,144	6.6%
Missing instrumental variable*	8,307	10.6%
Missing earnings information	1,450	1.8%
Missing covariates	1,227	1.6%
Never observed working in initial sector	5,168	6.6%
Final estimation sample	22,609	

Notes: This table reports sample restrictions applied to the initial population of PhD graduates observed in the 2015 Survey of Doctorate Recipients. Percentages are reported relative to the initial sample of 78,320 individuals. Health-related PhD majors refer to audiology and speech pathology, health services administration, medicine, nursing, pharmacy, physical therapy, public health and other health and medical sciences. *: because of less than 10 individuals in a major $\times$ cohort cell. The final estimation sample consists of individuals who satisfy all restrictions simultaneously.

Table B.2: OLS regression of Log(Earnings) on sector joined

	Log(Earnings)				
	(1)	(2)	(3)	(4)	(5)
Industry	0.482***	0.489***	0.414***	0.413***	0.398***
	(0.010)	(0.010)	(0.013)	(0.012)	(0.012)
PhD Cohort FE	Yes	Yes	Yes	Yes	Yes
Experience		Yes	Yes	Yes	Yes
PhD major FE			Yes	Yes	Yes
Doct. inst. FE				Yes	Yes
Demographics					Yes
Observations	69,914	69,914	69,914	69,914	69,914
R-sq	0.14	0.18	0.22	0.24	0.26

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The outcome is Log(Earnings). *Industry* is an indicator equal to 1 if the individual joins industry and 0 otherwise. *Experience* includes a linear and quadratic terms for experience, calculated as the difference between survey year and PhD graduation year. *Demographics* includes an indicator equal to 1 if the individual is female, an indicator equal to 1 if the individual is white, an indicator equal to 1 if at least one of the individual's parents has a doctoral degree, an indicator for being an American citizen and birth place fixed-effects. Standard errors are clustered at the doctoral institution level.

Table B.3: Falsification tests - Exclusion restriction

	Employer Size			Occupation		Log(Earnings)	
	Small	Medium	Large	R&D	Services	First	All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Z_i	0.021	-0.002	-0.020	0.011	0.017	0.079	0.057
	(0.066)	(0.079)	(0.071)	(0.063)	(0.054)	(0.104)	(0.076)
PhD major FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
PhD cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Experience	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Doct. inst. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,654	4,654	4,654	4,569	4,569	4,714	14,913
R-sq	0.22	0.18	0.20	0.33	0.32	0.21	0.20

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. Regressions are performed within the subsample of individuals who enter industry. Results are similar when considering the sample of individuals entering academia. The outcome in Column (1) is an indicator equal to 1 if the firm has less than 100 employees and 0 otherwise. The outcome in Column (2) is an indicator equal to 1 if the firm has between 100 and 24,999 employees and 0 otherwise. The outcome in Column (3) is an indicator equal to 1 if the firm has 25,000 or more employees and 0 otherwise. The outcome in Column (4) is an indicator equal to 1 if the individual's primary occupation upon PhD graduation is defined as Research and Development and 0 otherwise. The outcome in Column (5) is an indicator equal to 1 if the individual's primary occupation upon PhD graduation is defined as professional services to individuals and 0 otherwise. The outcome in Column (6) is the log of earnings for the earliest year of experience at which the individual is observed. The outcome in Column (7) is the log of earnings. *Demographics* includes an indicator equal to 1 if the individual is female, an indicator equal to 1 if the individual is white, an indicator equal to 1 if at least one of the individual's parents has a doctoral degree, an indicator for being an American citizen and birth place fixed-effects. *Experience* includes a linear and quadratic terms for experience, calculated as the difference between survey year and PhD graduation year. All regressions control for field-specific federal and private R&D funding at the time of graduation. Standard errors are clustered at the PhD major×PhD cohort level.

Table B.4: First-stage

	Industry (0/1)	
Z_i	0.179***	0.202***
	(0.041)	(0.044)
PhD major FE	Yes	Yes
PhD cohort	Yes	Yes
Experience	Yes	Yes
Demographics	Yes	Yes
Doct. Inst. FE	Yes	Yes
F-stat (Kleibergen-Paap)	20	21
Observations	22,608	69,914
R-sq	0.24	0.27

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The outcome is an indicator equal to 1 if individuals join industry and 0 otherwise. *Demographics* includes an indicator equal to 1 if the individual is female, an indicator equal to 1 if the individual is white, an indicator equal to 1 if at least one of the individual's parents has a doctoral degree, an indicator for being an American citizen and birth place fixed-effects. *Experience* includes a linear and quadratic terms for experience, calculated as the difference between survey year and PhD graduation year. All regressions control for field-specific federal and private R&D funding at the time of graduation. Standard errors are clustered at the PhD major$\times$PhD cohort level.

Table B.5: Alignment academic-industrial productivity and layoff rates, robustness

	Layoff$_{it}=1$		
	(1)	(2)	(3)
Alignment$_{major(i)}$	-0.025**	-0.074***	-0.023*
	(0.010)	(0.019)	(0.013)
Large Firm=1		-0.039***	
		(0.009)	
Alignment$_{major(i)}$ $\times$ Large Firm=1		0.101***	
		(0.024)	
R&D occupation=1			0.001
			(0.007)
Alignment$_{major(i)}$ $\times$ R&D occupation=1			-0.006
			(0.017)
PhD cohort FE	Yes	Yes	Yes
Survey year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
PhD field FE	Yes	Yes	Yes
Observations	10,258	6,601	9,963
Mean of dep. var.	0.01	0.01	0.01

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The dependent variable is an indicator equal to 1 if individual i reports not working in year t because of a layoff and 0 if individual i reports working. The independent variable is the correlation between individuals' productivity in academia and in industry, calculated at the PhD major level. Individuals' productivity in industry is proxied by their potential earnings in industry, residualized by PhD cohort, experience and experience squared, and averaged at the individual level. Individuals' productivity in academia is proxied by their potential citation-weighted publication output in academia (from graduation to 2017), residualized by PhD cohort, PhD major, and averaged at the individual level. Higher correlation reflects stronger alignment. Standard errors are clustered at the PhD major level.

Table B.6: Alignment academic-industrial productivity and separation outcomes (excl/ layoff)

Reason not working:	Family	Illness	Not want to work
	(1)	(2)	(3)
Alignment$_{major(i)}$	0.002	0.003	-0.000
	(0.008)	(0.003)	(0.005)
PhD cohort	Yes	Yes	Yes
Survey year	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Location	Yes	Yes	Yes
PhD field	Yes	Yes	Yes
Observations	10,246	10,219	10,228

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The dependent variable in Column (1) is an indicator equal to 1 if individual i reports not working in year t because of family reasons and 0 if individual i reports working. The dependent variable in Column (2) is an indicator equal to 1 if individual i reports not working in year t because of illness and 0 if individual i reports working. The dependent variable in Column (3) is an indicator equal to 1 if individual i reports not working in year t because of a preference not to work and 0 if individual i reports working. The independent variable is the correlation between individuals' productivity in academia and in industry, calculated at the PhD major level. Individuals' productivity in each sector is proxied by their potential earnings in each sector, residualized by PhD cohort, experience and experience squared, and averaged at the individual level. Higher correlation reflects stronger alignment. Standard errors are clustered at the PhD major level.

Table B.7: Alignment academic-industrial productivity and separation outcomes (excl/ layoff), robustness

Reason not working:	Family (1)	Illness (2)	Not want to work (3)
Alignment$_{major(i)}$	-0.013 (0.012)	0.001 (0.003)	-0.004 (0.007)
PhD cohort	Yes	Yes	Yes
Survey year	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Location	Yes	Yes	Yes
PhD field	Yes	Yes	Yes
Observations	10,230	10,203	10,212

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. The dependent variable in Column (1) is an indicator equal to 1 if individual i reports not working in year t because of family reasons and 0 if individual i reports working. The dependent variable in Column (2) is an indicator equal to 1 if individual i reports not working in year t because of illness and 0 if individual i reports working. The dependent variable in Column (3) is an indicator equal to 1 if individual i reports not working in year t because of a preference not to work and 0 if individual i reports working. The independent variable is the correlation between individuals' productivity in academia and in industry, calculated at the PhD major level. Individuals' productivity in industry is proxied by their potential earnings in industry, residualized by PhD cohort, experience and experience squared, and averaged at the individual level. Individuals' productivity in academia is proxied by their potential citation-weighted publication output in academia (from graduation to 2017), residualized by PhD cohort, PhD major, and averaged at the individual level. Higher correlation reflects stronger alignment. Standard errors are clustered at the PhD major level.

Table B.8: Academic signals at graduation and potential earnings in industry

	Log(Potential earnings in Industry)						
	Training & Publication					*Funding*	*Preferences*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Postdoc	-0.046***	-0.060***	-0.051***	-0.054***	-0.054***	-0.054***	-0.053***
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Top University		0.137***	0.136***	0.144***	0.144***	0.143***	0.142***
		(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Publications>0			0.001	0.001	0.001	0.001	-0.001
			(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Log(PhD duration)				-0.022***	-0.022***	-0.019***	-0.020***
				(0.007)	(0.007)	(0.007)	(0.007)
No Master					-0.013**	-0.013**	-0.011**
					(0.005)	(0.005)	(0.005)
Research Ass.						0.017***	0.014**
						(0.007)	(0.007)
Fellowship						0.007	0.005
						(0.007)	(0.007)
Grant						-0.002	-0.006
						(0.011)	(0.012)
Personal						-0.008	-0.010
						(0.009)	(0.010)
Employer						-0.001	-0.004
						(0.014)	(0.014)
Advancement							0.009*
							(0.005)
Benefits							-0.002
							(0.005)
Challenge							0.006
							(0.006)
Independence							-0.008
							(0.005)
Location							0.003
							(0.004)
Responsibility							0.008
							(0.005)
Salary							0.003
							(0.005)
Security							-0.011**
							(0.005)
Society							-0.012**
							(0.005)
PhD cohort	Yes	Yes	Yes	Yes	Yes	Yes	Yes
PhD major	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,826	21,826	18,180	13,589	13,589	13,241	12,120
Incr. R2 (baseline=0.71)	0.01	0.03	0.04	0.06	0.06	0.06	0.07

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. Standard errors are clustered at the PhD major level$\times$PhD cohort level.

Table B.9: Academic signals at graduation and potential earnings in academia

	Log(Potential earnings in Academia)						
	Training & Publication					*Funding*	*Preferences*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Postdoc	0.081***	0.068***	0.060***	0.062***	0.062***	0.062***	0.062***
	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Top University		0.133***	0.132***	0.128***	0.128***	0.126***	0.127***
		(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Publications>0			-0.004*	-0.003	-0.003	-0.004	-0.004
			(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Log(PhD duration)				0.002	0.002	0.003	0.003
				(0.004)	(0.004)	(0.004)	(0.004)
No Master					-0.002	-0.001	-0.001
					(0.002)	(0.002)	(0.003)
Research Ass.						0.009***	0.011***
						(0.003)	(0.003)
Fellowship						0.019***	0.018***
						(0.003)	(0.003)
Grant						0.003	0.003
						(0.005)	(0.005)
Personal						0.008*	0.007
						(0.004)	(0.005)
Employer						0.022**	0.024**
						(0.009)	(0.009)
Advancement							-0.007***
							(0.002)
Benefits							-0.001
							(0.003)
Challenge							0.003
							(0.003)
Independence							0.003
							(0.003)
Location							0.000
							(0.002)
Responsibility							-0.002
							(0.003)
Salary							-0.002
							(0.003)
Security							-0.002
							(0.003)
Society							0.010***
							(0.002)
PhD cohort	Yes	Yes	Yes	Yes	Yes	Yes	Yes
PhD major	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,826	21,826	18,180	13,589	13,589	13,241	12,120
Incr. R2 (baseline=0.50)	0.04	0.12	0.14	0.14	0.14	0.14	0.14

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. Standard errors are clustered at the PhD major level$\times$PhD cohort level.

Table B.10: Academic signals at graduation and potential citation-weighted publications in academia

	Log(Potential citation-weighted publications in Academia)						
	Training & Publication					*Funding*	*Preferences*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Postdoc	0.143***	0.086***	0.085***	0.085***	0.084***	0.065***	0.063***
	(0.009)	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)
Top University		0.638***	0.636***	0.641***	0.641***	0.622***	0.618***
		(0.007)	(0.007)	(0.009)	(0.009)	(0.009)	(0.009)
Publications>0			0.035***	0.034***	0.034***	0.025***	0.023**
			(0.008)	(0.008)	(0.008)	(0.008)	(0.009)
Log(PhD duration)				-0.017	-0.016	0.019	0.026*
				(0.014)	(0.014)	(0.015)	(0.016)
No Master=1					0.028***	0.022**	0.026***
					(0.009)	(0.009)	(0.010)
Research Ass.						0.098***	0.098***
						(0.012)	(0.012)
Fellowship						0.106***	0.104***
						(0.012)	(0.013)
Grant						0.088***	0.093***
						(0.018)	(0.018)
Personal						-0.058***	-0.060***
						(0.020)	(0.020)
Employer						-0.084***	-0.085***
						(0.031)	(0.032)
Advancement							0.005
							(0.008)
Benefits							-0.008
							(0.009)
Challenge							0.055***
							(0.009)
Independence							0.017*
							(0.009)
Location							0.019**
							(0.008)
Responsibility							-0.014
							(0.009)
Salary							-0.002
							(0.009)
Security							-0.027***
							(0.008)
Society							-0.002
							(0.008)
PhD cohort	Yes	Yes	Yes	Yes	Yes	Yes	Yes
PhD major	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,372	21,372	21,372	15,687	15,687	15,284	14,100
Incr. R2 (baseline=0.25)	0.01	0.23	0.23	0.23	0.24	0.24	0.25

Notes: *,**,*** denote significance at 10%, 5% and 1% level respectively. Standard errors are clustered at the PhD major level$\times$PhD cohort level.

Appendix C Modeling the Correlation between Academic and Industrial Productivity

Let each individual i have three skills: k_i^G is a general skill valued by both industry and academia, k_i^A is a skill valued only in academia and k_i^I is a skill valued only in industry. Assume productivity is linear in skills so that:

$$P_i^A = \theta_G^A k_i^G + \theta_A k_i^A$$
$$P_i^I = \theta_G^I k_i^G + \theta_I k_i^I$$

where θ_G^I denotes the return to general skills in industry, θ_G^A denotes the return to general skills in academia, θ_A denotes the return to academia-specific skills and θ_I denotes the return to industry-specific skills. For simplicity, assume $\theta_G^I = \theta_G^A = \theta_G$. The correlation between sector-specific productivity is given by:

$$\text{Corr}(P_i^A, P_i^I) = \frac{\text{Cov}(P_i^A, P_i^I)}{\sqrt{\text{Var}(P_i^A)} \cdot \sqrt{\text{Var}(P_i^I)}}$$

Sign. The sign of the correlation is driven by the sign of the numerator, which expands as follows:

$$\text{Cov}(P_i^A, P_i^I) = \theta_G^2 \cdot \text{Var}(k_i^G) + \theta_G\theta_I \cdot \text{Cov}(k_i^G, k_i^I) + \theta_G\theta_A \cdot \text{Cov}(k_i^G, k_i^A) + \theta_A\theta_I \cdot \text{Cov}(k_i^A, k_i^I)$$

Each term captures the following:

- $\theta_G^2 \cdot \text{Var}(k_i^G)$: Importance of shared general skills (always ≥ 0)
- $\theta_G\theta_I \cdot \text{Cov}(k_i^G, k_i^I)$: Complementarity between general and industry-specific skills, i.e., do individuals endowed with general skills also tend to be endowed with industry-specific skills?
- $\theta_G\theta_A \cdot \text{Cov}(k_i^G, k_i^A)$: Complementarity between general and academia-specific skills, i.e., do individuals endowed with general skills also tend to be endowed with academia-specific skills?
- $\theta_A\theta_I \cdot \text{Cov}(k_i^A, k_i^I)$: Do sector-specific skills co-occur (positive) or trade off (negative) within individuals?

The first term, which reflects the contribution of general skills, is always positive. It pushes the correlation up when general skills are both important (high θ_G) and variable (high $\text{Var}(k_i^G)$). The second and third terms capture whether individuals with strong general skills also tend to have strong sector-specific skills. If general and sector-specific skills positively co-occur within individuals, these terms are positive and increase correlation. However, if general skills are orthogonal or inversely related to one sector's skill, these terms reduce the correlation. The final term captures whether sector-specific skills co-occur or trade-off within individuals. If academia-specific and industry-specific skills trade-off within individuals, then $\text{Cov}(k_i^A, k_i^I)$ is negative, and the last term pulls the correlation down. This means that even if general skills are valued, the overall correlation can be low or negative if sector-specific skills are negatively related.

Magnitude. The overall magnitude of the correlation also depends on the denominator, which captures the standard deviations of productivity in each sector. The variances are given by:

$$\text{Var}(P_i^A) = \theta_G^2 \cdot \text{Var}(k_i^G) + \theta_A^2 \cdot \text{Var}(k_i^A) + 2\theta_G\theta_A \cdot \text{Cov}(k_i^G, k_i^A)$$
$$\text{Var}(P_i^I) = \theta_G^2 \cdot \text{Var}(k_i^G) + \theta_I^2 \cdot \text{Var}(k_i^I) + 2\theta_G\theta_I \cdot \text{Cov}(k_i^G, k_i^I)$$

An increase in the denominator can dilute the correlation when the variation in each sector is driven by components that don't co-move across sectors (i.e., when the numerator does not increase at the same time). This occurs, for instance, when productivity in each sector is heavily influenced by sector-specific skills that vary widely across individuals. In this case, the denominator increases through $\theta_A^2 \cdot \text{Var}(k_i^A)$ and $\theta_I^2 \cdot \text{Var}(k_i^I)$. If sector-specific skills are positively correlated ($\text{Cov}(k_i^A, k_i^I) > 0$), the correlation remains positive but is diluted. If they are negatively correlated ($\text{Cov}(k_i^A, k_i^I) < 0$), the correlation is pulled toward zero.

Special cases.

- If only general skills matter (i.e., there is no sector-specific skills), then:

$$\text{Corr}(P_i^A, P_i^I) = 1$$

- If only sector-specific skills matter, then:

$$\text{Corr}(P_i^A, P_i^I) = \frac{\theta_A\theta_I \cdot \text{Cov}(k_i^A, k_i^I)}{\sqrt{\text{Var}(P_i^A)} \cdot \sqrt{\text{Var}(P_i^I)}} = \text{Corr}(k_i^A, k_i^I)$$

Appendix D Selection on gains vs selection on level

I detail here the difference between selection on level and selection on gains and how they might be correlated with each other.

The utility function of individual i in sector j is $U_i^j = \beta_i W_i^j + \alpha_i 1\{j = \text{industry}\}$, where α_i captures individual i's preference for industry relative to academia and β_i reflects sensitivity to earnings. W_i^j can be decomposed into $W_i^j = \overline{W^j} + \delta_i^j$ where $\overline{W^j}$ captures the part of earnings explained by observables and δ_i^j captures the part of earnings related to unobservable components. Individual i joins industry iff $\beta_i(\overline{W^I} - \overline{W^A}) + \beta_i(\delta_i^I - \delta_i^A) + \alpha_i > 0$.

The first selection mechanism is selection on *gains* which refers to the value of the difference $\delta_i^I - \delta_i^A$ and influences which sector individuals choose at graduation. Individuals with higher values of $\delta_i^I - \delta_i^A$ have more to gain by going to the private sector and are thus more likely to select into that sector. The second selection mechanism refers to selection on *level* which is linked to the values of δ_i^I and δ_i^A. Consider two individuals identical on observable characteristics and with the same gains $\delta_i^I - \delta_i^A$, but assume that individual 1 has higher level values of δ_i^I and δ_i^A compared to individual 2. Because earnings are the sum of a sector-mean (common to individuals 1 and 2) and an individual-specific component (higher for individual 1), individual 1's potential earning outcomes are higher than individual 2's potential earning outcomes. While the sector selection equation is only linked to $\delta_i^I - \delta_i^A$, this difference might be correlated with the levels δ_i^I and δ_i^A except if we assume constant treatment effects. To see that, let's derive the conditions under which the difference and the levels are uncorrelated:

$$
\begin{gathered}
Cov(\delta_i^I - \delta_i^A, \delta_i^I) = 0 \\
Cov(\delta_i^I - \delta_i^A, \delta_i^A) = 0 \\
\iff \\
Var(\delta_i^I) - Cov(\delta_i^A, \delta_i^I) = 0 \\
Cov(\delta_i^I, \delta_i^A) - Var(\delta_i^A) = 0 \\
\iff \\
Cov(\delta_i^A, \delta_i^I) = Var(\delta_i^I) = Var(\delta_i^A) \\
\iff \\
\frac{Cov(\delta_i^A, \delta_i^I)}{\sqrt{Var(\delta_i^I)Var(\delta_i^A)}} = \frac{Var(\delta_i^I)}{\sqrt{Var(\delta_i^I)Var(\delta_i^A)}} = 1
\end{gathered}
$$

This implies a linear relationship between δ_i^I and δ_i^A:

$$\delta_i^I = a + b\delta_i^A$$

with a and b some constant. Plugging back into the initial conditions:

$$Cov(\delta_i^I - \delta_i^A, \delta_i^I) = 0$$
$$Cov(\delta_i^I - \delta_i^A, \delta_i^A) = 0$$
$$\Longleftrightarrow$$
$$Cov(a + b\delta_i^A - \delta_i^A, \delta_i^I) = 0$$
$$Cov(a + b\delta_i^A - \delta_i^A, \delta_i^A) = 0$$
$$\Longleftrightarrow$$
$$Cov(a + (b-1)\delta_i^A, \delta_i^I) = 0$$
$$Cov(a + (b-1)\delta_i^A, \delta_i^A) = 0$$
$$\Longleftrightarrow$$
$$(b-1)Cov(\delta_i^A, \delta_i^I) = 0$$
$$(b-1)Var(\delta_i^A) = 0$$

which implies that $b = 1$ or δ_i^I and δ_i^A being constant. In both cases, this implies that the difference $\delta_i^I - \delta_i^A$ is a constant.

Appendix E Marginal Treatment Effects (MTE)

The Marginal Treatment Effects (MTE) framework provides a natural way to study heterogeneity in the returns to treatment (in this paper, this would be working in industry versus academia). The key idea is that individuals differ both in how much they would benefit from working in industry and in their propensity to enter the private sector. Some individuals are eager to join industry and would do so under almost any circumstances, while others are more reluctant and would only enter if strongly pushed (by e.g., labor market conditions).

The MTE formalizes this intuition by ordering individuals along a continuum based on their unobserved propensity to enter industry. At each point along this continuum, the MTE captures the treatment effect of entering industry for individuals with that level of resistance to the private sector. Tracing out the MTE therefore reveals how the returns to industry vary across individuals with different propensities to sort into industry.

Importantly, familiar treatment effect parameters, such as the Local Average Treatment Effect (LATE), the Average Treatment Effect (ATE), and the Average Treatment Effect on the Treated (ATT), can all be understood as weighted averages of this underlying MTE curve, with different estimands placing weight on different parts of the distribution. Moreover, when using a separate approach to estimate MTE (Grennan et al., 2024), one can also recover potential outcomes rather than only treatment effects. When productivity proxies are used as outcomes, this approach allows me to recover empirical counterparts to the key objects of interest in the paper: individuals' productivity in academia, P_i^A, and in industry, P_i^I.

Formally, MTE can be modeled as part of the generalized Roy Model (see Andresen (2018) for a more extensive discussion). I now formalize this framework for my empirical setting.

Model – I follow the generalized Roy model and assume that each individual has two potential outcomes, one in each sector:

$$P_i^j = \overline{P^j}(X_i) + \nu_i^j, \quad j \in \{I, A\}$$

where $\overline{P^j}(X_i)$ captures the component of productivity explained by observable characteristics X_i, and ν_i^j captures unobserved determinants of productivity in sector j. P_i^A denotes individual i's productivity in academia, P_i^I denotes individual i's productivity in industry. Sector choice is governed by the following selection equation:

$$Industry_i = \mathbf{1}\{P(Z_i, X_i) > U_{D_i}\}$$

where $Industry_i$ equals 1 if individual i enters industry and 0 otherwise. $P(X_i, Z_i)$ is the propensity score, i.e., the probability of entering industry based on observed covariates X_i and an instrument Z_i. U_{D_i} is an unobserved resistance often conceptualized as a distaste for working in industry, normalized to be uniformly distributed between 0 and 1. Individuals with *lower* values of U_D are *more inclined* to enter industry than those with *higher* values.

At any given value of the propensity score $p_i = P(X_i, Z_i)$, individuals observed in industry must have $U_{D_i} \leq p_i$, while those observed in academia must have $U_{D_i} > p_i$. Specifying some function for the conditional expectations of the error terms allows me to use the observed outcomes of individuals in industry and the observed outcomes of individuals in academia to estimate the following conditional expectations of P_i^A and P_i^I:

- $E[P_i^I \mid X_i{=}x_i,\, p_i,\, Industry_i{=}1] = x_i\beta_1 + E[\nu_i^I \mid U_{D_i} \leq p_i] \equiv x_i\beta_1 + K_1(p_i)$
- $E[P_i^A \mid X_i{=}x_i,\, p_i,\, Industry_i{=}0] = x_i\beta_0 + E[\nu_i^A \mid U_{D_i} \geq p_i] \equiv x_i\beta_0 + K_0(p_i)$

Based on the functional form assumptions chosen for $K_1(p)$ and $K_0(p)$, we can recover marginal potential outcomes, i.e., expected productivity at a specific value of unobserved resistance:

- $E[P_i^I \mid X_i{=}x_i,\, U_{D_i}{=}u] = x_i\beta_1 + E[\nu_i^I \mid U_{D_i}{=}u]$
- $E[P_i^A \mid X_i{=}x_i,\, U_{D_i}{=}u] = x_i\beta_0 + E[\nu_i^A \mid U_{D_i}{=}u]$.

Taking the difference yields the Marginal Treatment Effect (MTE):

$$MTE(x,u) = E[P_i^I - P_i^A | X_i = x, U_{D_i} = u]$$

The MTE represents the treatment effect for individuals with observable characteristics x and latent resistance level u. It captures heterogeneity in treatment effects arising from both observed and unobserved factors.

Predictions - If I knew the unobservable U_{D_i} for each individual, I could easily infer their potential outcomes in academia, P_i^A, and in industry, P_i^I, based on the previous estimations. While U_{D_i} is unobserved, I know for each individual i: i) the sector that they actually join and ii) their propensity score p_i. For each individual i with a propensity score p_i observed in industry, we know that they have $U_{D_i} \leq p_i$. For each individual i with a propensity score p_i observed in academia, we know that they have $U_{D_i} \geq p_i$. Hence:

$$\mathbf{E}[P_i^I | x_i, p_i, Industry_i = 1] = x_i\beta_1 + E[\nu_i^I | U_{D_i} \leq p_i]$$

$$\mathbf{E}[P_i^I | x_i, p_i, Industry_i = 0] = x_i\beta_1 + E[\nu_i^I | U_{D_i} \geq p_i]$$

which gives:

$$\mathbf{E}[P_i^I | x_i, p_i, Industry_i] = x_i\beta_1 + \frac{Industry_i - p_i}{1 - p_i} K_1(p_i) \tag{3}$$

Similarly we can derive:

$$E[P_i^A \mid X_i{=}x_i,\, p_i,\, Industry_i] = x_i\beta_0 + \frac{p_i - Industry_i}{p_i} K_0(p_i) \tag{4}$$

Equation 3 represents individual i's expected productivity in industry. Equation 4 represents individual i's expected productivity in academia.

Assumptions - The assumptions needed for the calculation of the MTE are the same as for the LATE: relevance, exclusion and monotonicity. In theory, it is possible to estimate MTE with no further assumption. However, this requires full support of the propensity score in both the treated and untreated states for all values of X which is rarely feasible (Andresen, 2018). I follow the literature and further assume additive separability between the observed and unobserved components (Carneiro et al., 2011; Cornelissen et al., 2016). While this MTE framework is more restrictive than the TSLS one, it allows me to bring more nuance into treatment heterogeneity and the pattern of selection.[26] MTE can be estimated through the local IV or the separate approach. I use the latter because it allows me to estimate potential outcomes, which are my main parameters of interest.

[26]Note that the separability assumption remains much less restrictive than a joint normal distribution of (U^A, U^I, U_D) as assumed by traditional selection models (Andresen, 2018).